Jobs to end poverty?

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What role should human capital play in poverty reduction strategies?

- At a macro-level, human capital likely had a key role in global poverty reduction.
  - 60pct of 1980-2019 income growth among bottom 20pct can be accounted for by human capital accumulation (Gethin 2025).
  - Unclear whether additional human capital would have high returns.
  - We will focus on the case of training.
- At a micro level, training raises employment by 5ppt and earnings by 11 to 24 pct (Agarwal and Mani 2024).
  - Selection into experimental samples is likely not random.
  - Typically, training is fully subsidized.
  - Meaningful heterogeneity.

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$\rightarrow$ Unclear what role should training have in our poverty reduction toolkit.

Key questions

- What are the returns to training among the poor?
- Can the poor afford to pay for the full cost of training? Or are subsidies necessary?
- How heterogeneous are the returns to training? Do people know they own returns? Can selection be improved?

This project

1. We study take-up and impacts of randomized discounts offered to attend Prosper, a training program (OJT) run by BRAC in Bangladesh.
   - Prior to intervention, participants pay full cost of the program.
   - We interview a representative sample of potential trainees and randomize the offer of training at different price points.

2. We use ML methods to test whether the patterns we observe in Bangladesh are found in other existing studies.

3. We compare BRAC's selection to selection in the experiment/selection at scale.
   - BRAC officers actively recruit trainees (selection + persuasion)
   - What is BRAC's objective and does BRAC have useful information on returns?

4. We calibrate a model to quantify the (i) returns to training for different groups, (ii) the uncertainty on these returns, and (iii) whether a scale up is possible.

Contributions

1. We better identify who benefits from training, with a particular focus on the poor.
   Kluve et al. 2019, Agarwal and Mani 2023, Carranza and McKenzie 2024

2. We explore the implications for targeting and scale-up.
   Kitagawa Tetenov 2018, Caria et al. 2023, Haushofer et al. 2025, Bjorkegren et al. 2025
   - Test for an impact-deprivation tradeoff for human capital interventions.
   - Identify how benefit uncertainty shapes selection and the potential for scale-up.

Roadmap

A simple framework

The impact of training among the poor
- Design
- Results

How general are our results?

Can BRAC improve targeting?
- Design
- Results

Training to end poverty?

A simple model of selection into training

The net benefit from training is $B - (1 + C)pd$.

- A heterogeneous monetary benefit from training B.
- A monetary cost p. We offer discounts $d \in \{1, .7, .3\}$.
- An additional contingent cost C (e.g., excess cost of liquidity).
  - $C \in \{c_{poor}, c_{rich}\}$

- C, p, d are observed, but B is not.
- Risk-neutral potential trainees observe a signal of B: $\widehat{B}_1 = B + E1$
- BRAC observes an unrelated signal: $\widehat{B}_2 = B + E2$

The staus quo and the experiment

BRAC selection:

- *Selectivity* $\rightarrow$ BRAC invites a share q with the highest expected take-up $Pr[B - (1 + C)pd \geq 0 | \widehat{B}_2 = \widehat{b}_2]$
- *Persuasion* $\rightarrow$ Invitees observe $\widehat{B}_2$ and enrol if: $E[B | \widehat{B}_1 = \widehat{b}_1, \widehat{B}_2 = \widehat{b}_2] - (1 + C)pd \geq 0$

Experiment selection:

- Invite a random sample of individuals interested in training
- They will enrol if: $E[B | \widehat{B}_1 = \widehat{b}_1] - (1 + C)pd \geq 0$

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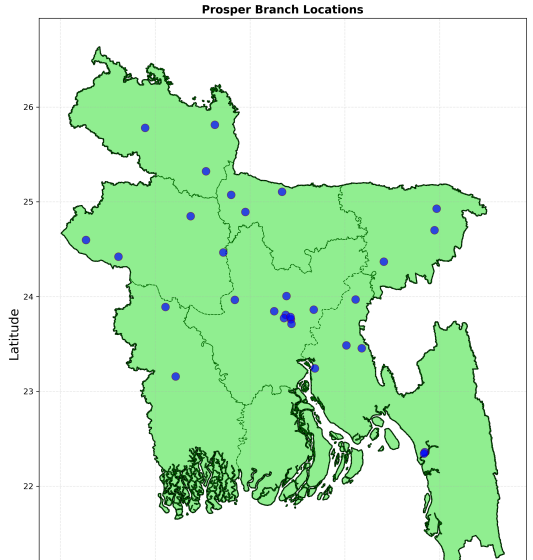
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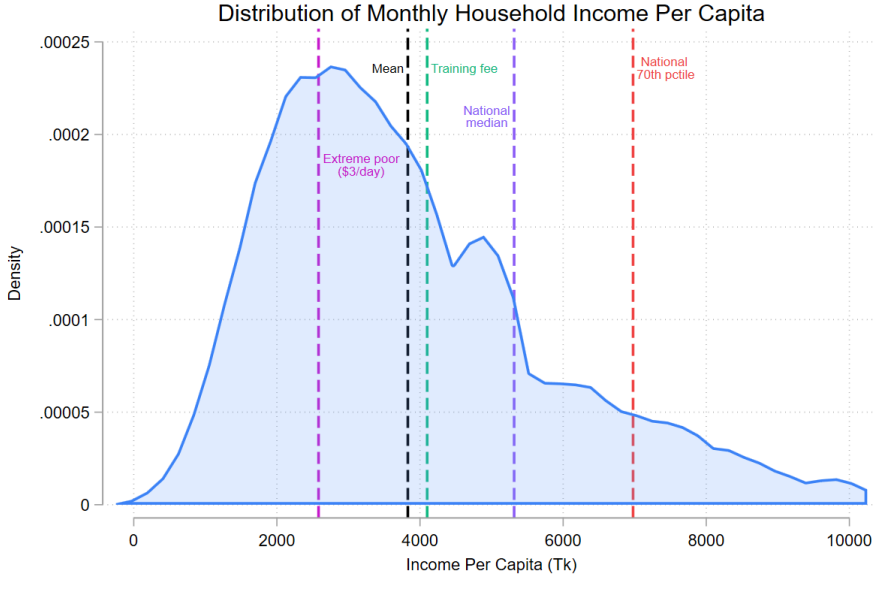
Training to end poverty?

The Prosper program

- Operates in urban/peri-urban areas
- 3 months unpaid internship with master craft person
- 4 sectors: Graphic design, IT repair, Tailoring, Beauty Specialist
- Cost: 4100Tk



The distribution of income



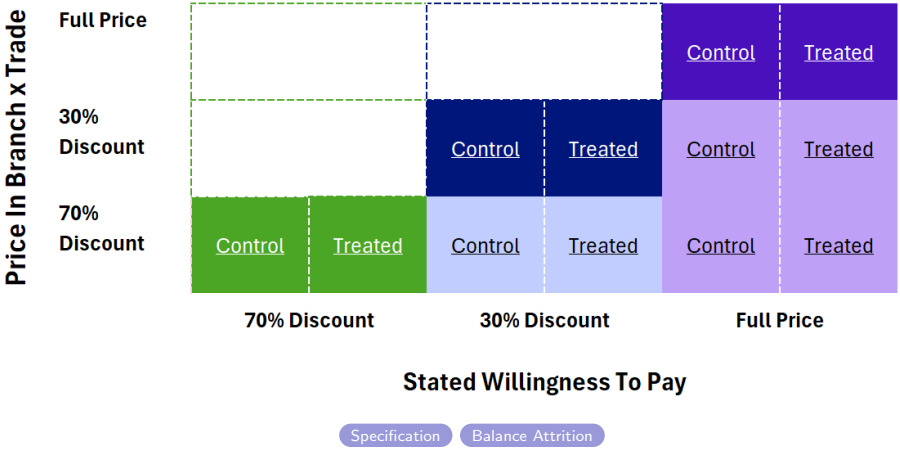
Income deciles

Full sample vs. interested

Sampling

- We collect a census of individuals living in randomly selected neighborhoods close to BRAC's training centers in peri-urban Bangladesh (N=42,895)
- Eligibility: persons age 16-40 living in households with monthly incomes below 10,000 Tk per capita (N=37,389)
- We further restrict to those interested in at least 1 of the 4 trades (N=18,901)
- We ask individuals to rank their interest in the 4 trades and whether they would be interested in training in their preferred trade at the following conditions:
  1. Full price
  2. 30 pct discount
  3. 70 pct discount

At the branch x trade level, we randomize whether the training is offered at full price, 30pct discount, 70pct discount.



Specification | Balance Attrition

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Training to end poverty?

Enrollment and completion increase sharply with the subsidy

Table: Effect of poverty on enrollment

	Enrolled (1)	Completed (2)
Full price	0.035**	0.027**
	(0.014)	(0.012)
Discount 30pct	0.057***	0.043***
	(0.012)	(0.009)
Discount 70pct	0.146***	0.134***
	(0.014)	(0.013)
p-value for equality of treatment effects:		
	0.000	0.000
Control mean	0.000	0.000
Observations	4,744	4,744
R-squared	0.105	0.100

Standard errors are clustered by branch-trade.
*** p<0.01, ** p<0.05, * p<0.1

Intensive and extensive margins

Subsidies attract poorer individuals

Table: Impact of discount on trainee composition

	HH income per capita		Poverty (2021 PPP)	
	Interested (1)	Enrolled (2)	Interested (3)	Enrolled (4)
Constant	4280.352	4289.364	0.238	0.273
	(94.627)	(519.243)	(0.018)	(0.106)
30% discount	-284.293***	-241.544	0.019	0.011
	(86.948)	(556.890)	(0.017)	(0.125)
70% discount	-523.606***	-725.354	0.079***	0.046
	(73.930)	(543.899)	(0.015)	(0.113)
R^2	0.014	0.021	0.007	0.002
Observations	10583	322	10583	322

Odd columns use the interested sample with maximum-WTP dummies. Even columns use the enrolled sample with enrolment-price dummies (omitted category is full price in both). Poverty is an indicator for HH income per capita below \$3/day (2021 PPP). SE clustered at branch-trade. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Takeup effects by poverty

No average impacts on labor market outcomes

Table: Treatment Effects on Labor Market Outcomes

	Employed	Hours	Earnings
	(1)	(2)	(3)
Full price	-0.040	-0.597	-253.138
	(0.028)	(0.898)	(275.482)
Discount 30pct	-0.020	-1.489	-403.123
	(0.025)	(1.348)	(310.166)
Discount 70pct	0.015	1.000	163.655
	(0.018)	(0.693)	(171.010)
p-value for equality of treatment effects:			
	0.217	0.168	0.206
Control mean	0.355	11.592	3117.483
Observations	4,246	4,246	4,246
R-squared	0.200	0.303	0.389
Rerandomized Controls	Yes	Yes	Yes

Standard errors are clustered by branch-trade. All regressions include `ib1.ctreat` (Full price as the omitted category) and rerandomized balance controls. Earnings are measured in 1000 Tk. Control mean is unweighted (`invite==0`). *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

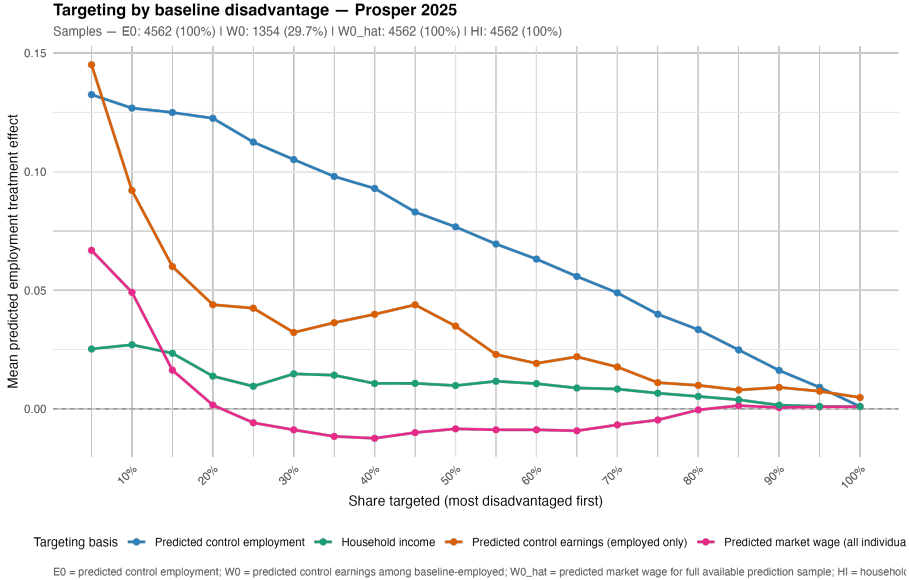
Highly subsidised training improves outcomes for the poor

Table: Treatment Effects on Labor Market Outcomes

	Enrolment (1)	Employment (2)	Earnings (3)	Indiv. Food (4)	HH Non-Food (5)
Panel A: Below $3.00/day					
Full price	0.033	-0.021	-295.8	625.9***	-2139.7
	(0.028)	(0.056)	(574.8)	(222.7)	(1589.0)
Discount 30pct	0.067***	-0.085	-471.0	-235.1	160.2
	(0.025)	(0.064)	(635.5)	(227.3)	(1277.5)
Discount 70pct	0.162***	0.090**	242.3	240.0*	1737.3*
	(0.034)	(0.040)	(422.6)	(130.0)	(962.8)
Control mean	0.000	0.332	2659.2	3654.1	16035.8
Observations	1,074	1,074	1,074	1,074	1,074
R-squared	0.104	0.221	0.367	0.131	0.056
Panel B: Above $3.00/day					
Full price	0.032*	-0.044	-219.9	-34.8	5.7
	(0.018)	(0.037)	(313.1)	(118.9)	(887.6)
Discount 30pct	0.059***	-0.001	-443.8	198.9	-282.7
	(0.015)	(0.029)	(374.5)	(158.3)	(706.3)
Discount 70pct	0.137***	-0.016	140.0	24.9	643.8
	(0.021)	(0.024)	(209.0)	(122.2)	(1275.9)
Control mean	0.000	0.363	3266.5	4289.8	18700.7
Observations	3,172	3,172	3,172	3,172	3,172
R-squared	0.074	0.202	0.402	0.081	0.063
Rerandomized Controls	Yes	Yes	Yes	Yes	Yes

Sample split by baseline household income per capita relative to the $3.00/day (2021 PPP) extreme poverty line. Standard errors are clustered by branch-trade. Control variables are rerandomized balance controls, a dummy variable for baseline student status, preferred trade, baseline earnings, self-employed status at baseline, self-employed status in Nov. 2019, and aspired earnings. Earnings are measured in Tk. Individual food expenditure is the monthly equivalent of the response to: In the past 7 days, how many days have you eaten the following food items listed? How much would it cost to buy all of the food you consumed from this category this week at a grocery store in taka? Household non-food expenditure is the sum of monthly expenditure on non-food non-durables and the annual expenditure on durables divided by 12. Earnings and consumption variables are winsorized at the 99th percentile. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

If we target on need, returns are higher than average: utilitarian and Rawlsian targeting rules coincide



Roadmap

A simple framework

The impact of training among the poor

How general are our results?

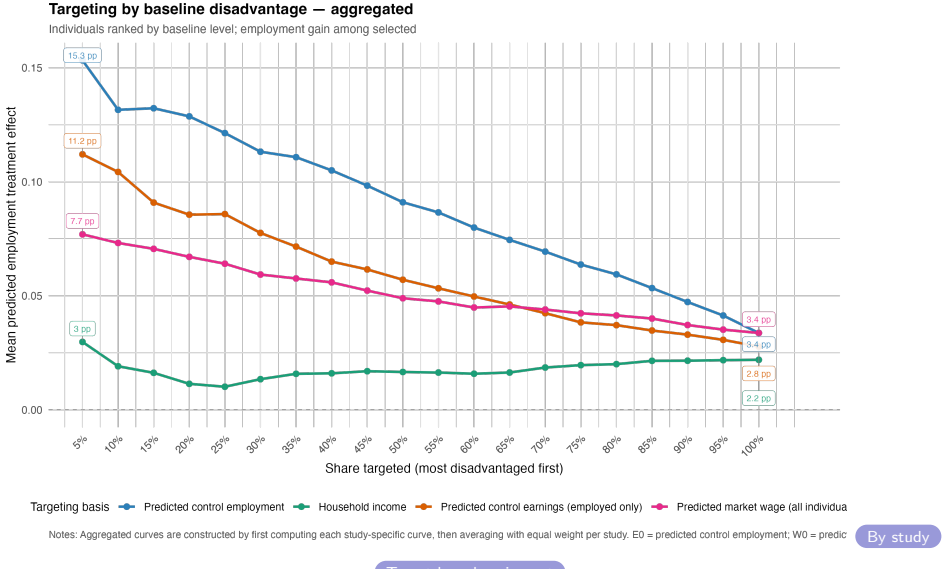
Can BRAC improve targeting?

Training to end poverty?

- We re-analyse all studies in Agarwal and Mani (2024) that have micro-data available (10 studies) A&M Map Reanalysis Map
  - **Training is offered for free** to treated individuals in all of these studies;
  - **Recruitment protocols involve significant pre-selection** Recruitment

- We use the ML method of Chernozukhov et al. (2022), to predict, for every individual, a $\hat{Y}_0$ and a $\hat{Y}_1$

If we target on need, returns are average to higher-than-average



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BRAC targeting

- Shortly after the end of the experiment, BRAC ran a new wave of Prosper.
- **They selected participants as usual (selection + persuasion)**. Time Use Recruit
- Minimal overlap with self-selection experiment, as they focused on different close-by neighborhoods than those we had (randomly) chosen for the experiment. Map
- No pure control group.
- A hypothetical recruitment exercise with BRAC selectors confirms they prioritise people who are likely to enrol. Hypothetical recruitment

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Training to end poverty?

BRAC targets better off individuals...

Table: Trainee composition by sample

	HH income pc	Poverty (2021 PPP)
	(1)	(2)
Constant	4688.941	0.237
	(571.160)	(0.059)
BRAC	1171.018***	-0.066
	(347.174)	(0.057)
R^2	0.043	0.013
Observations	1419	1419

Sample restricted to enrolled respondents in the census and BRAC samples at baseline. Each column regresses the outcome on an indicator for the BRAC sample (omitted category is the census enrolled sample), controlling for trade and discount size. Poverty is an indicator for HH income per capita below $3/day (2021 PPP). SE clustered at branch-trade level. * p<0.1, ** p<0.05, *** p<0.01.

Demographics Full price

... who are more likely to take-up the program

Table: Treatment Effects on Training Enrollment and Completion by Price

	Unweighted		Weighted	
	Enrolled	Completed	Enrolled	Completed
	(1)	(2)	(3)	(4)
Full price	0.035**	0.027*	0.045***	0.033***
	(0.016)	(0.014)	(0.014)	(0.012)
Discount 30pct	0.060***	0.045***	0.068***	0.047***
	(0.013)	(0.010)	(0.015)	(0.011)
Discount 70pct	0.143***	0.131***	0.175***	0.161***
	(0.015)	(0.014)	(0.023)	(0.021)
p-value for equality of treatment effects:				
	0.000	0.000	0.000	0.000
Control mean	0.000	0.000	0.000	0.000
Observations	4,050	4,050	4,050	4,050
R-squared	0.101	0.096	0.118	0.113

Standard errors are clustered by branch-trade.
*** p<0.01, ** p<0.05, * p<0.1

... and to benefit from it

Table: Employment Treatment Effects

	Unweighted (1)	Variable Wts (2)	Takeup Wts (3)
Full price	-0.040	0.001	-0.016
	(0.028)	(0.048)	(0.043)
Discount 30pct	-0.020	-0.034	-0.034
	(0.025)	(0.029)	(0.028)
Discount 70pct	0.015	0.063*	0.053*
	(0.018)	(0.033)	(0.031)
p-value for equality of treatment effects:			
	0.217	0.091	0.108
Control mean	0.355	0.355	0.355
Observations	4,246	4,246	4,246
R-squared	0.200	0.246	0.240
Rerandomized Controls	Yes	Yes	Yes

Outcome is an indicator for employment at endline. Standard errors are clustered by branch-trade. All regressions include `ib1.ctreat` (Full price as the omitted category) and rerandomized balance controls. Control mean is unweighted (`invite==0`). *** p$¡$0.01, ** p$¡$0.05, * p$¡$0.1.

▸ By poverty status

Implications

- BRAC likely favors people with $C = c_{rich}$, as they are more likely to take-up
- But also has some genuine information on B.

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Training to end poverty?

The basic model has reasonable fit (more moments to be added)

Table: Simulated and targeted moments

Moment	Targeted (data)	Simulated (model)
Pr[enrol | d=1.0, non-poor]	0.032	0.000
Pr[enrol | d=1.0, poor]	0.033	0.000
Pr[enrol | d=0.7, non-poor]	0.059	0.000
Pr[enrol | d=0.7, poor]	0.067	0.000
Pr[enrol | d=0.3, non-poor]	0.137	0.097
Pr[enrol | d=0.3, poor]	0.162	0.140
E[B | enrol, d=0.3, non-poor]	1021.7	1527.6
E[B | enrol, d=0.3, poor]	1495.5	2242.7
Pr[enrol | inv$_{d=1}$, d=0.3, non-poor]	0.157	0.188
Pr[enrol | inv$_{d=1}$, d=0.3, poor]	0.255	0.280
E[B | inv$_{d=1}$, enrol, d=0.3, non-poor]	2873.0	2843.9
E[B | inv$_{d=1}$, enrol, d=0.3, poor]	5299.5	4649.5
Value of loss function		4.660

And parameter estimates point to the fact that returns are higher among the poor, but also highly heterogeneous

Table: Calibrated parameters

	Non-poor c_l	Poor c_h
Mean benefit μ_B	622.1	796.0
Benefit std. dev. σ_B	2541.2	3493.1
Excess cost c	0.039	0.465
Worker signal std. dev. σ_{E_w}	12479.2	
NGO signal std. dev. σ_{E_n}	1656.4	
NGO invitation share q	0.442	
Signal-transmission prob. x	0.099	

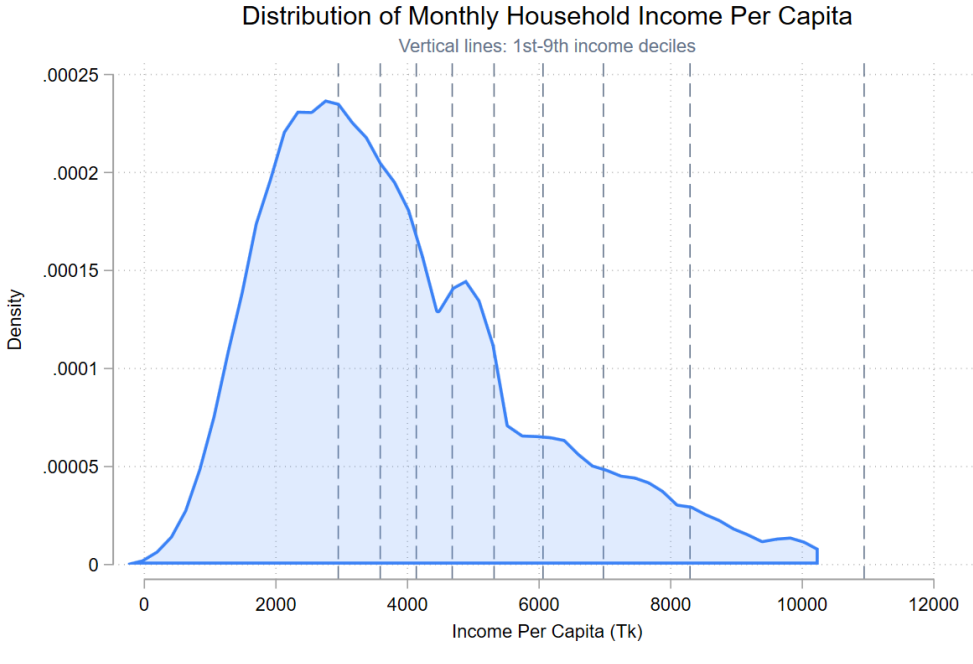
To conclude

- Our results (so far) suggest there are high returns to training for *some* of the poor.
- The utilitarian and Rawlsian targeting rules coincide, but NGOs have incentives and information contraints that can lead to deviations from these rules.

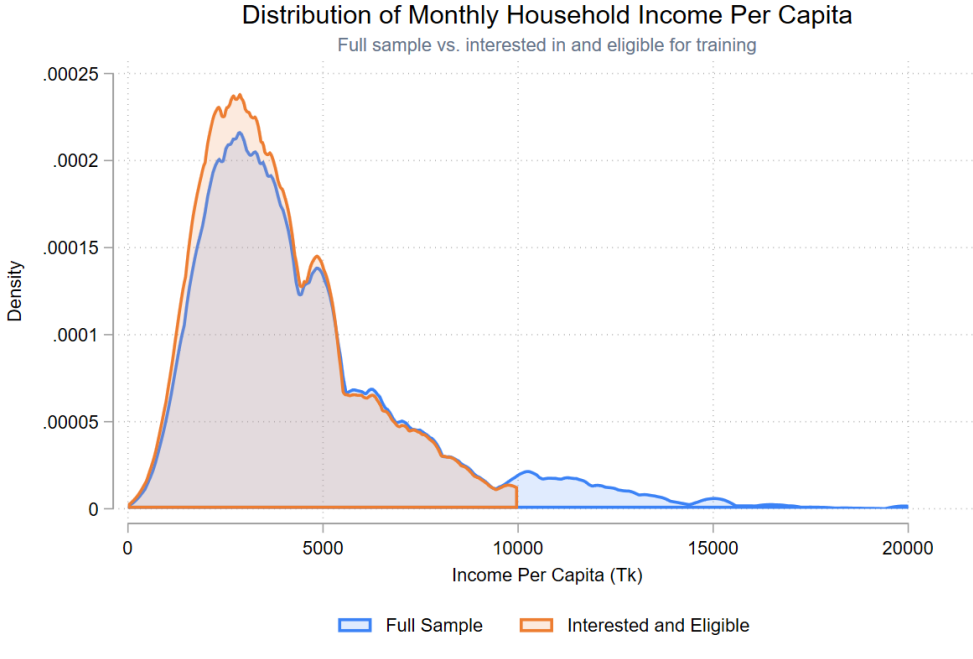
$\rightarrow$ Targeted human capital interventions can be part of poverty reduction strategy.

- Key next step: understand the factors that generate the heterogeneity we observe.

Thank You!


Distribution of Monthly Household Income Per Capita
Vertical lines: 1st-9th income deciles
Density
.00025
.0002
.00015
.0001
.00005
0
0
2000
4000
6000
8000
10000
12000
Income Per Capita (Tk)

Back


Distribution of Monthly Household Income Per Capita
Full sample vs. interested in and eligible for training
.00025
.0002
.00015
.0001
.00005
0
Density
0
5000
10000
15000
20000
Income Per Capita (Tk)
Full Sample
Interested and Eligible

Back

Table: Balance by Baseline Characteristics (Part 1)

	Full price		30% discount		70% discount			
Variable	Control	Treat	Control	Treat	Control	Treat	Joint p	N
Female	0.78	0.74	0.64	0.67	0.79	0.84	0.14	5623
	(0.42)	(0.44)	(0.48)	(0.47)	(0.41)	(0.37)		
Age	24.18	24.91	23.82	23.96	25.52	25.40	0.06*	5625
	(6.42)	(6.78)	(6.43)	(6.40)	(7.09)	(6.84)		
Migrant	0.41	0.41	0.30	0.33	0.33	0.31	0.47	5625
	(0.49)	(0.49)	(0.46)	(0.47)	(0.47)	(0.46)		
Num children	0.95	0.99	0.77	0.82	1.00	1.13	0.05**	5625
	(1.08)	(1.10)	(1.13)	(1.04)	(1.12)	(1.11)		
Age youngest child	3.66	4.22	3.73	3.66	4.77	4.25	0.01**	3101
	(3.90)	(4.01)	(3.44)	(3.61)	(4.23)	(3.98)		
HH head	0.09	0.10	0.09	0.09	0.10	0.09	0.83	5625
	(0.29)	(0.30)	(0.29)	(0.28)	(0.30)	(0.28)		
Secondary educ	0.19	0.21	0.24	0.21	0.21	0.21	0.95	5625
	(0.40)	(0.40)	(0.43)	(0.41)	(0.41)	(0.40)		
Any work experience	0.45	0.47	0.48	0.52	0.45	0.44	0.21	5625
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)		
In work	0.25	0.29	0.32	0.30	0.25	0.24	0.15	5625
	(0.43)	(0.46)	(0.47)	(0.46)	(0.43)	(0.43)		
Ever trained	0.20	0.22	0.25	0.22	0.26	0.22	0.19	5625
	(0.40)	(0.41)	(0.44)	(0.41)	(0.44)	(0.41)		
Has liquidity	0.42	0.39	0.31	0.34	0.22	0.25	0.17	5625
	(0.49)	(0.49)	(0.46)	(0.47)	(0.41)	(0.43)		
Large expected wage gain	0.49	0.50	0.49	0.52	0.44	0.44	0.37	3164
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)		
Joint F-test: $p = 0.29$								

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Table: Balance by Baseline Characteristics (Part 2)

	Full price		30% discount		70% discount			
Variable	Control	Treat	Control	Treat	Control	Treat	Joint p	N
Student	0.33	0.29	0.42	0.37	0.27	0.25	0.23	5625
	(0.47)	(0.45)	(0.49)	(0.48)	(0.45)	(0.43)		
Self-employed	0.08	0.08	0.10	0.11	0.10	0.12	0.51	5625
	(0.28)	(0.28)	(0.30)	(0.31)	(0.30)	(0.32)		
Self-employed (Nov 2019)	0.09	0.09	0.10	0.10	0.08	0.10	0.29	5625
	(0.29)	(0.29)	(0.30)	(0.30)	(0.27)	(0.30)		
Earnings	1.36	2.09	2.12	1.88	1.43	1.17	0.05**	5625
	(3.54)	(4.74)	(4.57)	(4.45)	(3.84)	(3.49)		
Earnings (Nov 2019)	2.04	2.65	2.56	2.35	1.75	1.75	0.24	5625
	(5.08)	(5.88)	(5.72)	(5.22)	(4.37)	(4.61)		
Aspired earnings	17.17	15.86	18.11	17.69	14.33	13.52	0.60	5625
	(29.31)	(21.28)	(16.15)	(16.16)	(14.95)	(12.86)		
Expected job search length change	-1.93	-1.77	-2.43	-2.28	-1.72	-2.04	0.29	3709
	(3.35)	(2.94)	(3.64)	(3.12)	(2.83)	(2.98)		
Endline attrition	0.24	0.27	0.24	0.25	0.24	0.23	0.41	5625
	(0.43)	(0.45)	(0.43)	(0.43)	(0.42)	(0.42)		
Predicted endline employment	0.33	0.35	0.38	0.37	0.33	0.31	0.03**	5625
	(0.18)	(0.20)	(0.20)	(0.19)	(0.18)	(0.17)		

Joint F-test: $p = 0.29$

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Back

Table: Balance by Baseline Characteristics - Full Price (Part 1)

	Full price			
Variable	Control	Treat	p-value	N
Female	0.78	0.74	0.06*	1616
	(0.42)	(0.44)		
Age	24.18	24.91	0.02**	1616
	(6.42)	(6.78)		
Migrant	0.41	0.41	0.59	1616
	(0.49)	(0.49)		
Num children	0.95	0.99	0.28	1616
	(1.08)	(1.10)		
Age youngest child	3.66	4.22	0.26	883
	(3.90)	(4.01)		
HH head	0.09	0.10	0.54	1616
	(0.29)	(0.30)		
Secondary educ	0.19	0.21	0.80	1616
	(0.40)	(0.40)		
Any work experience	0.45	0.47	0.18	1616
	(0.50)	(0.50)		
In work	0.25	0.29	0.04**	1616
	(0.43)	(0.46)		
Ever trained	0.20	0.22	0.59	1616
	(0.40)	(0.41)		
Has liquidity	0.42	0.39	0.32	1616
	(0.49)	(0.49)		
Large expected wage gain	0.49	0.50	0.67	863
	(0.50)	(0.50)		
Joint F-test: p = 0.00***				

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Table: Balance by Baseline Characteristics - Full Price (Part 2)

	Full price			
Variable	Control	Treat	p-value	N
Student	0.33 (0.47)	0.29 (0.45)	0.07*	1616
Self-employed	0.08 (0.28)	0.08 (0.28)	0.99	1616
Self-employed (Nov 2019)	0.09 (0.29)	0.09 (0.29)	0.95	1616
Earnings	1.36 (3.54)	2.09 (4.74)	0.02**	1616
Earnings (Nov 2019)	2.04 (5.08)	2.65 (5.88)	0.09*	1616
Aspired earnings	17.17 (29.31)	15.86 (21.28)	0.34	1616
Expected job search length change	-1.93 (3.35)	-1.77 (2.94)	0.33	1071
Endline attrition	0.24 (0.43)	0.27 (0.45)	0.14	1616
Predicted endline employment	0.33 (0.18)	0.35 (0.20)	0.01**	1616
Joint F-test: p = 0.00***				

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Back

Table: Balance by Baseline Characteristics - Discount 30 pct (Part 1)

	30% discount			
Variable	Control	Treat	p-value	N
Female	0.64	0.67	0.97	1530
	(0.48)	(0.47)		
Age	23.82	23.96	0.48	1531
	(6.43)	(6.40)		
Migrant	0.30	0.33	0.59	1531
	(0.46)	(0.47)		
Num children	0.77	0.82	0.56	1531
	(1.13)	(1.04)		
Age youngest child	3.73	3.66	0.61	702
	(3.44)	(3.61)		
HH head	0.09	0.09	0.96	1531
	(0.29)	(0.28)		
Secondary educ	0.24	0.21	0.72	1531
	(0.43)	(0.41)		
Any work experience	0.48	0.52	0.24	1531
	(0.50)	(0.50)		
In work	0.32	0.30	0.47	1531
	(0.47)	(0.46)		
Ever trained	0.25	0.22	0.36	1531
	(0.44)	(0.41)		
Has liquidity	0.31	0.34	0.16	1531
	(0.46)	(0.47)		
Large expected wage gain	0.49	0.52	0.06*	923
	(0.50)	(0.50)		
Joint F-test: p = 0.00***				

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Table: Balance by Baseline Characteristics - Discount 30pct (Part 2)

	30% discount			
Variable	Control	Treat	p-value	N
Student	0.42	0.37	0.56	1531
	(0.49)	(0.48)		
Self-employed	0.10	0.11	0.79	1531
	(0.30)	(0.31)		
Self-employed (Nov 2019)	0.10	0.10	0.92	1531
	(0.30)	(0.30)		
Earnings	2.12	1.88	0.53	1531
	(4.57)	(4.45)		
Earnings (Nov 2019)	2.56	2.35	0.70	1531
	(5.72)	(5.22)		
Aspired earnings	18.11	17.69	0.28	1531
	(16.15)	(16.16)		
Expected job search length change	-2.43	-2.28	0.62	1049
	(3.64)	(3.12)		
Endline attrition	0.24	0.25	0.92	1531
	(0.43)	(0.43)		
Predicted endline employment	0.38	0.37	0.87	1531
	(0.20)	(0.19)		
Joint F-test: p = 0.00***				

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Back

Table: Balance by Baseline Characteristics - Discount 70pct (Part 1)

	70% discount			
Variable	Control	Treat	p-value	N
Female	0.79	0.84	0.16	2477
	(0.41)	(0.37)		
Age	25.52	25.40	0.24	2478
	(7.09)	(6.84)		
Migrant	0.33	0.31	0.28	2478
	(0.47)	(0.46)		
Num children	1.00	1.13	0.01**	2478
	(1.12)	(1.11)		
Age youngest child	4.77	4.25	0.00***	1516
	(4.23)	(3.98)		
HH head	0.10	0.09	0.62	2478
	(0.30)	(0.28)		
Secondary educ	0.21	0.21	0.97	2478
	(0.41)	(0.40)		
Any work experience	0.45	0.44	0.66	2478
	(0.50)	(0.50)		
In work	0.25	0.24	0.91	2478
	(0.43)	(0.43)		
Ever trained	0.26	0.22	0.06*	2478
	(0.44)	(0.41)		
Has liquidity	0.22	0.25	0.10*	2478
	(0.41)	(0.43)		
Large expected wage gain	0.44	0.44	0.91	1378
	(0.50)	(0.50)		
Joint F-test: p = 0.00***				

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * p<0.1, ** p<0.05, *** p<0.01.

Table: Balance by Baseline Characteristics - Discount 70pct (Part 2)

	70% discount			
Variable	Control	Treat	p-value	N
Student	0.27	0.25	0.63	2478
	(0.45)	(0.43)		
Self-employed	0.10	0.12	0.16	2478
	(0.30)	(0.32)		
Self-employed (Nov 2019)	0.08	0.10	0.07*	2478
	(0.27)	(0.30)		
Earnings	1.43	1.17	0.26	2478
	(3.84)	(3.49)		
Earnings (Nov 2019)	1.75	1.75	0.35	2478
	(4.37)	(4.61)		
Aspired earnings	14.33	13.52	0.68	2478
	(14.95)	(12.86)		
Expected job search length change	-1.72	-2.04	0.17	1589
	(2.83)	(2.98)		
Endline attrition	0.24	0.23	0.90	2478
	(0.42)	(0.42)		
Predicted endline employment	0.33	0.31	0.16	2478
	(0.18)	(0.17)		
Joint F-test: p = 0.00***				

Earnings variables reflect monthly earnings measured in 1000 Taka. Aspired earnings refer to aspired earnings 5 years in the future. Large expected wage gain is an indicator for whether the respondent has an above-median expected monthly wage increase from training. Has liquidity is an indicator for whether the respondent would be able to pay the full cost of training out of pocket this month. Expected job search length change is the expected change in the number of months required to find a job due to training. Predicted endline employment is the LASSO-linear prediction trained on baseline characteristics of the control group (age, education, household income, gender, children, employment, work experience, vocational training). * p<0.1, ** p<0.05, *** p<0.01.

Back

Table: Treatment Effects by Price (no Demand Type Controls)

	Unweighted			Weighted		
	Employed	Work hrs	Earnings	Employed	Work hrs	Earnings
	(1)	(2)	(3)	(4)	(5)	(6)
Full price	0.044**	2.430***	650.277***	0.076***	3.831***	646.598*
	(0.020)	(0.732)	(222.349)	(0.029)	(0.985)	(335.002)
Discount 30pct	-0.009	0.794	130.826	-0.018	0.601	-408.386
	(0.022)	(0.839)	(199.067)	(0.027)	(1.024)	(350.554)
Discount 70pct	0.006	0.818	116.150	0.082***	4.568***	841.987**
	(0.013)	(0.588)	(167.813)	(0.029)	(0.952)	(346.780)
Pay if employed	-0.017	-0.842	-456.879**	-0.012	-0.395	-653.018**
	(0.017)	(0.682)	(177.557)	(0.026)	(0.908)	(255.441)
Endline control means by treatment:						
Full price	0.376	10.500	2711.955	0.403	9.841	3303.121
Discount 30pct	0.409	13.238	3735.820	0.426	12.976	4117.964
Discount 70pct	0.290	8.549	2156.150	0.413	13.308	3884.266
Pay if employed	0.337	8.765	2202.736	0.361	10.710	3033.444
Observations	6,802	6,802	6,802	6,802	6,802	6,802
R-squared	0.066	0.088	0.121	0.051	0.073	0.085

Standard errors are clustered by branch-trade. The dependent variables are an indicator for whether the respondent is currently employed in a salary/wage-based job (columns 1 and 4), average weekly work hours over the past year (2 and 5) and average monthly earnings over the past year (3 and 6).
*** p<0.01, ** p<0.05, * p<0.1

BRAC targets better off individuals: demographics

Table: Trainee demographics by sample

	Male	No children	Ever married	Age	Trained	Any work exp.	Employed
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Constant	0.647	0.879	0.177	20.774	0.172	0.503	0.288
	(0.110)	(0.114)	(0.115)	(1.521)	(0.072)	(0.098)	(0.079)
BRAC	0.188***	0.212***	-0.185***	-0.776	-0.014	0.024	0.044
	(0.054)	(0.056)	(0.058)	(0.846)	(0.059)	(0.061)	(0.041)
R^2	0.100	0.119	0.090	0.029	0.001	0.005	0.004
Observations	1419	1419	1419	1419	1419	1419	1419

Sample restricted to enrolled respondents in the census and BRAC samples at baseline. Each column regresses the outcome on an indicator for the BRAC sample (omitted category is the census enrolled sample), controlling for trade and discount size. SE clustered at branch-trade level. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Back

Measuring household income per capita

During the baseline survey, a respondent is asked to provide their household's total income in the last month. This is then divided by the number of people living in the household to arrive at per capita household income.

BRAC targets better off individuals: full price

Table: Trainee composition by sample - full price households

	HH income pc	Poverty (2021 PPP)
	(1)	(2)
Constant	4976.291	0.216
	(1275.539)	(0.126)
BRAC	1467.442*	-0.097
	(790.743)	(0.129)
R^2	0.014	0.008
Observations	406	406

Sample restricted to respondents enrolled in the census and BRAC samples under full price. Each column regresses the outcome on an indicator for the BRAC sample (omitted category is the census enrolled sample), controlling for trade. Poverty is an indicator for HH income per capita below $3/day (2021 PPP). SE clustered at branch-trade level. * p<0.1, ** p<0.05, *** p<0.01.

Back

Table: Pooled Treatment Effects (with re-randomisation variable controls)

	Unweighted			Weighted		
	Employed	Work hrs	Earnings	Employed	Work hrs	Earnings
	(1)	(2)	(3)	(4)	(5)	(6)
Invited	-0.006	0.218	-69.960	0.011	1.184*	-51.484
	(0.011)	(0.418)	(105.322)	(0.018)	(0.684)	(220.622)
Control mean	0.344	9.749	2532.997	0.385	11.037	3295.039
Observations	6,800	6,800	6,800	6,800	6,800	6,800
R-squared	0.195	0.261	0.317	0.178	0.239	0.259

Standard errors are clustered by branch-trade. The dependent variables are an indicator for whether the respondent is currently employed in a salary/wage-based job (columns 1 and 4), average weekly work hours over the past year (2 and 5) and average monthly earnings over the past year (3 and 6). Variables used during re-randomization have been included as controls.

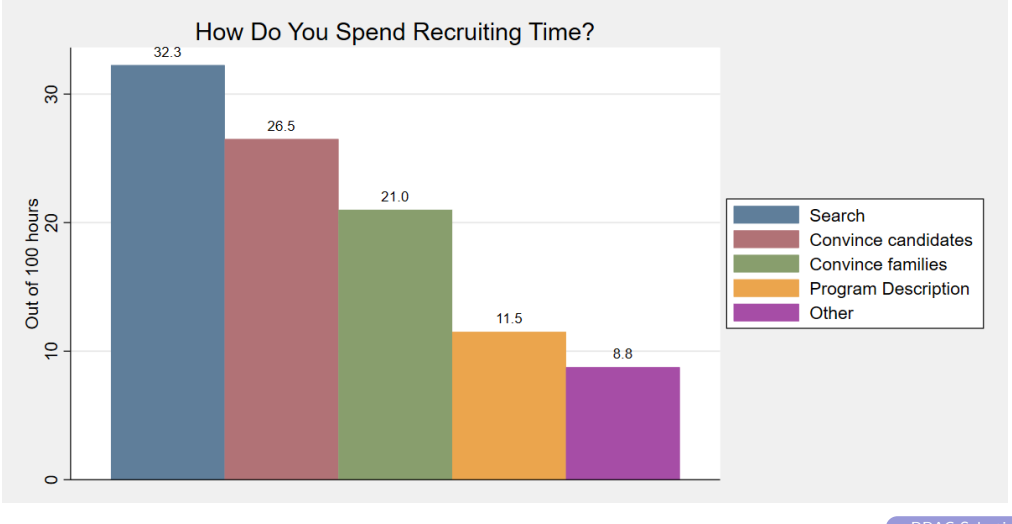
*** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table: Treatment Effects by Price (with re-randomisation variable controls)

	Unweighted			Weighted		
	Employed	Work hrs	Earnings	Employed	Work hrs	Earnings
	(1)	(2)	(3)	(4)	(5)	(6)
Full price	0.023	1.845***	455.933**	0.052*	3.418***	508.129
	(0.019)	(0.651)	(202.590)	(0.028)	(0.812)	(333.745)
Discount 30pct	-0.036	-0.319	-203.387	-0.045	-0.098	-628.621
	(0.022)	(0.956)	(205.352)	(0.027)	(1.301)	(389.159)
Discount 70pct	0.014	1.082*	202.453	0.075**	3.924***	720.424**
	(0.011)	(0.549)	(135.327)	(0.030)	(0.957)	(352.994)
Pay if employed	-0.020	-0.810	-413.863***	0.001	0.051	-459.395**
	(0.017)	(0.612)	(142.426)	(0.024)	(0.837)	(223.247)
p-value for equality of treatment effects:						
	0.011	0.001	0.000	0.001	0.000	0.000
Endline control means by treatment:						
Full price	0.376	10.500	2711.955	0.403	9.841	3303.121
Discount 30pct	0.409	13.238	3735.820	0.426	12.976	4117.964
Discount 70pct	0.290	8.549	2156.150	0.413	13.308	3884.266
Pay if employed	0.337	8.765	2202.736	0.361	10.710	3033.444
Observations	6,800	6,800	6,800	6,800	6,800	6,800
R-squared	0.197	0.263	0.320	0.194	0.253	0.276

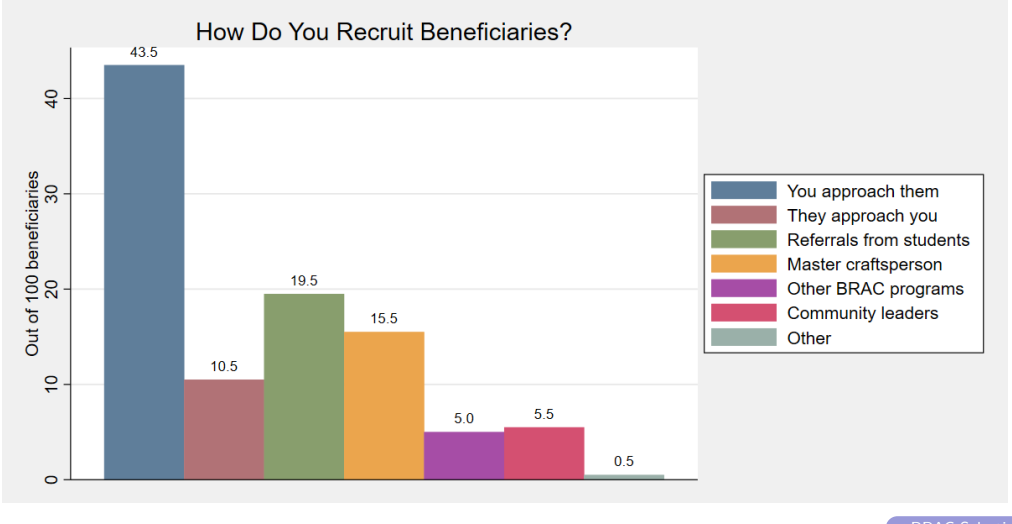
Standard errors are clustered by branch-trade. The dependent variables are an indicator for whether the respondent is currently employed in a salary/wage-based job (columns 1 and 4), average weekly work hours over the past year (2 and 5) and average monthly earnings over the past year (3 and 6). Variables used during re-randomization have been included as controls.
*** p<0.01, ** p<0.05, * p<0.1

Time Use



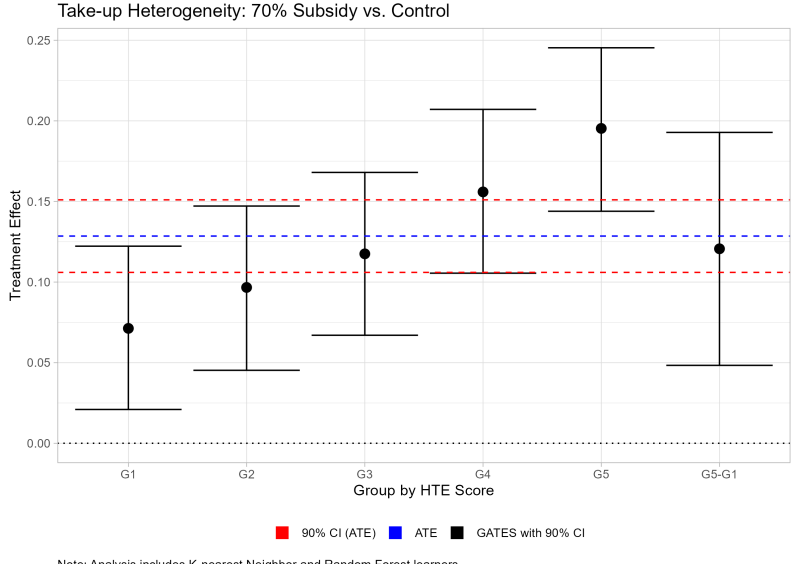
BRAC Selection

Recruiting Activities



BRAC Selection

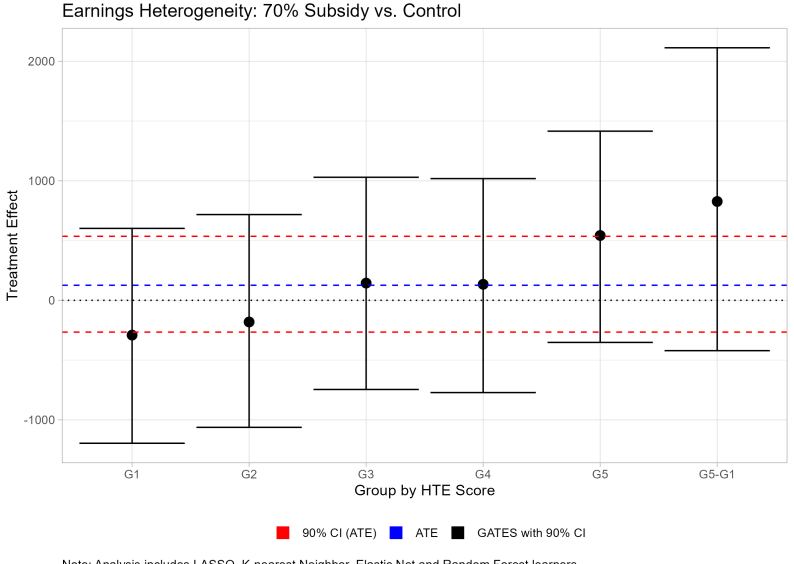
Takeup Heterogeneity



Note: Analysis includes K-nearest Neighbor and Random Forest learners
with 100 sample splits. Using demand controls, rerandomization variables, CV variables

Employment

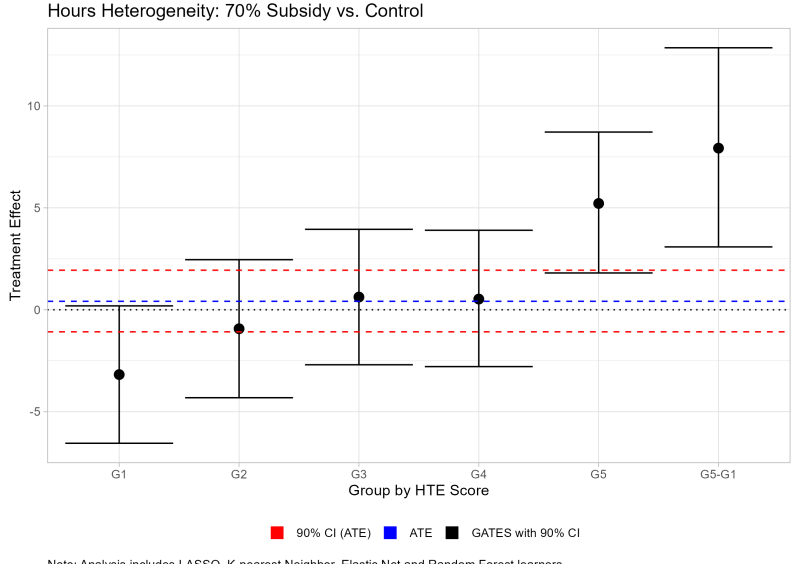
Earnings Heterogeneity



Note: Analysis includes LASSO, K-nearest Neighbor, Elastic Net and Random Forest learners with 100 sample splits. Using demand controls, rerandomization variables, CV variables

Employment

Hours Heterogeneity



Note: Analysis includes LASSO, K-nearest Neighbor, Elastic Net and Random Forest learners with 100 sample splits. Using demand controls, rerandomization variables, CV variables

Employment

Classification Analysis - Take-up

Table: Quintile 5 - Quintile 1 Differences: Take-up

Female	Age	Ever Married	Children	Education
-0.348***	-3.486***	-0.540***	-0.858***	0.107***
(0.032)	(0.550)	(0.032)	(0.085)	(0.032)
Father Skill	**Mother Skill**	**Migrant**	**Vocational Training**	**Liquidity**
-0.179***	0.068**	-0.118***	0.182***	-0.092***
(0.027)	(0.031)	(0.036)	(0.028)	(0.034)

Notes: * p<0.1, ** p<0.05, *** p<0.01. Standard errors in parentheses.

Employment

Classification Analysis - Earnings

Table: Quintile 5 - Quintile 1 Differences: Earnings

Age	Ever Married	Children	Work Experience	HH Head
-4.344***	-0.221***	-0.509***	-82.179***	-0.134***
(0.516)	(0.038)	(0.091)	(3.770)	(0.025)

Vocational Training	Employed
-0.134***	-0.742***
(0.031)	(0.025)

Notes: * p<0.1, ** p<0.05, *** p<0.01. Standard errors in parentheses.

Employment

Classification Analysis - Hours

Table: Quintile 5 - Quintile 1 Differences: Hours

Female	Age	Ever Married	Children	Education
-0.704***	-7.063***	-0.514***	-1.110***	-0.560***
(0.026)	(0.510)	(0.034)	(0.082)	(0.034)
Vocational Training	Work Experience	Employed		
-0.121***	-75.022***	-0.560***		
(0.032)	(3.902)	(0.034)		

Notes: * p<0.1, ** p<0.05, *** p<0.01. Standard errors in parentheses.

◂ Employment

Completion rates close to 1, except in PIE

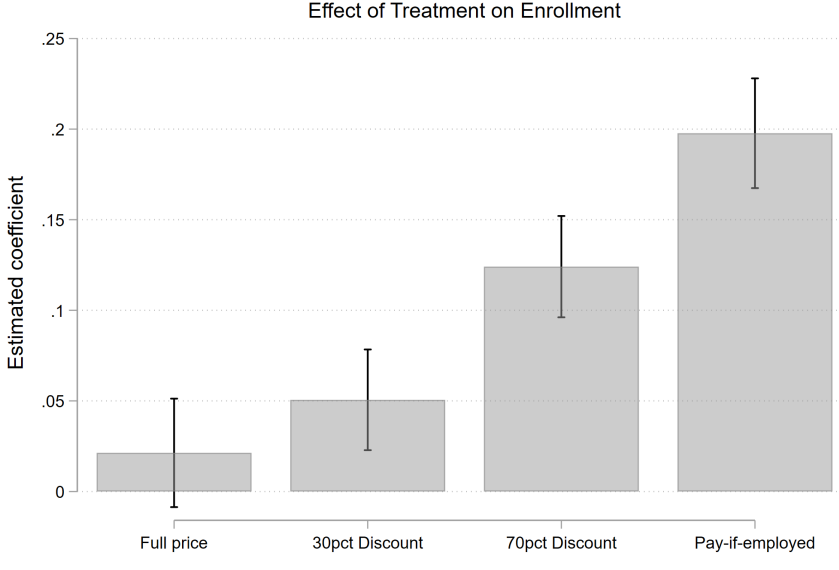
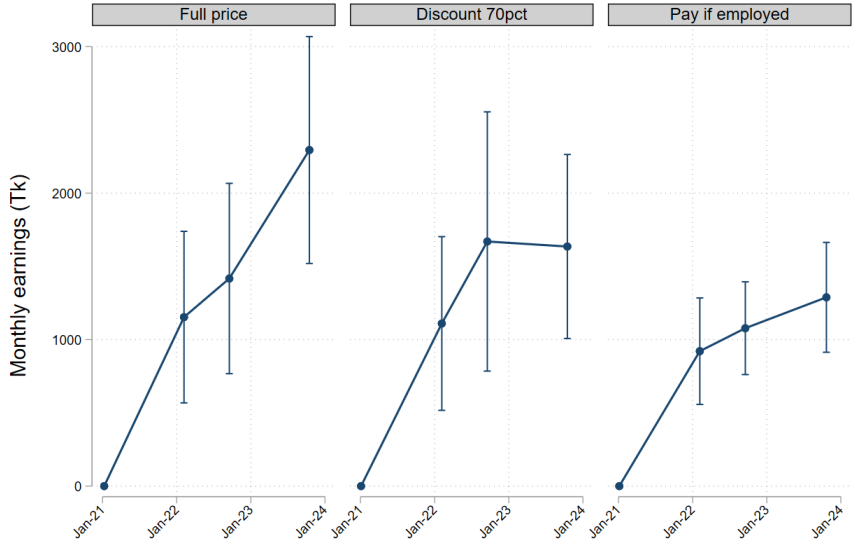


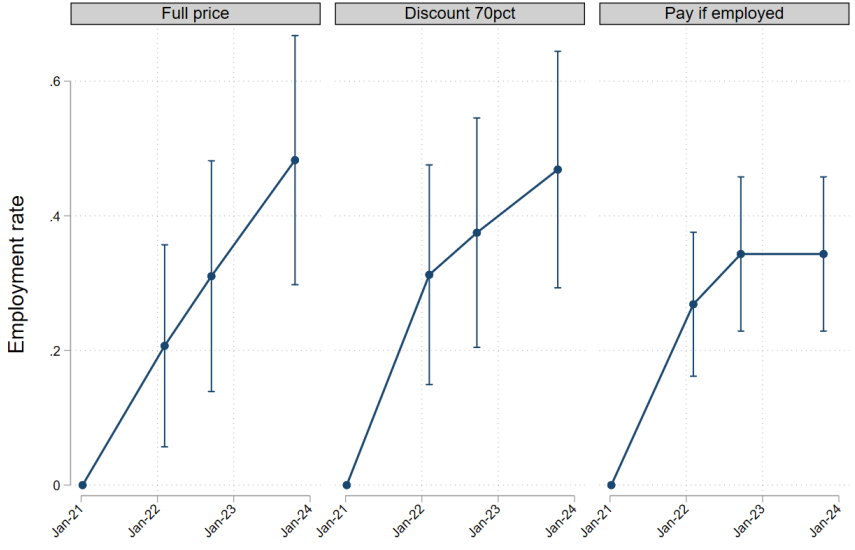
Table Back

Figure: Average Earnings in the Control Group - Unemployed at Baseline



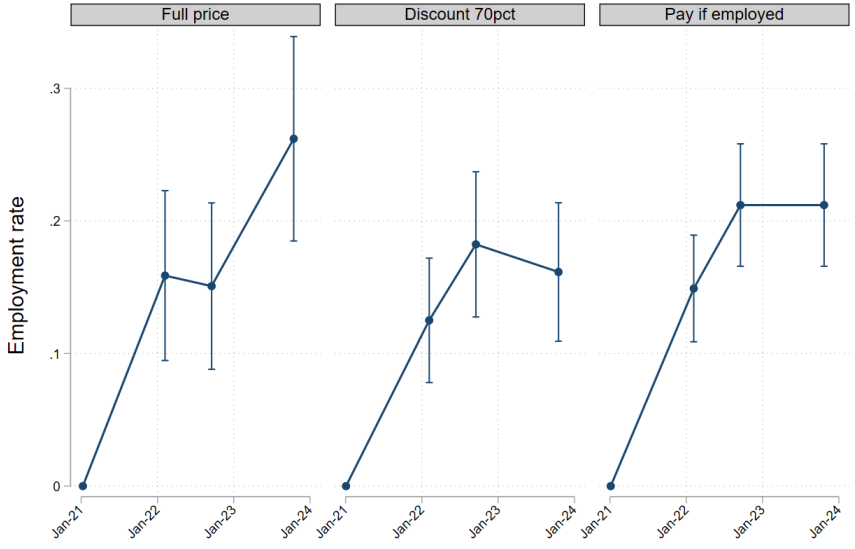
Back

Figure: Employment in the Control Group - Men Unemployed at Baseline



Back

Figure: Employment in the Control Group - Women Unemployed at Baseline



Back

Figure: Average Earnings in the Control Group - Men Unemployed at Baseline

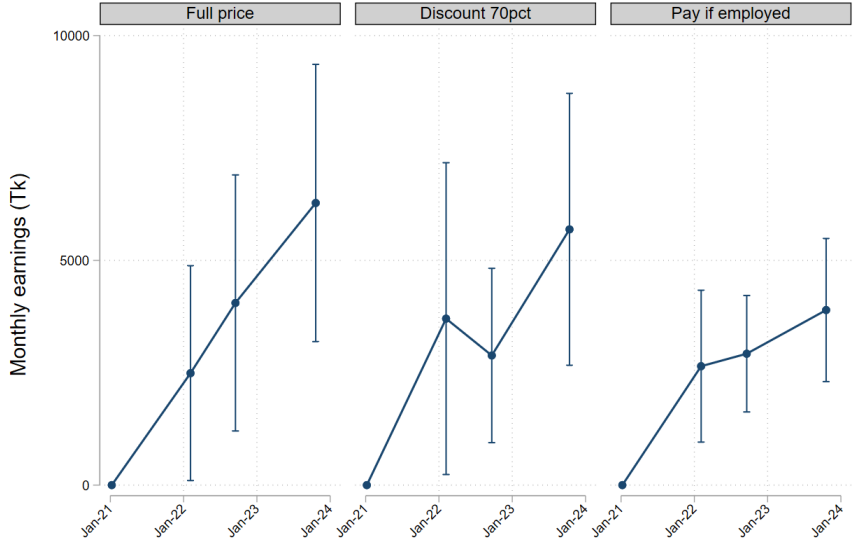


Figure: Average Earnings in the Control Group - Women Unemployed at Baseline

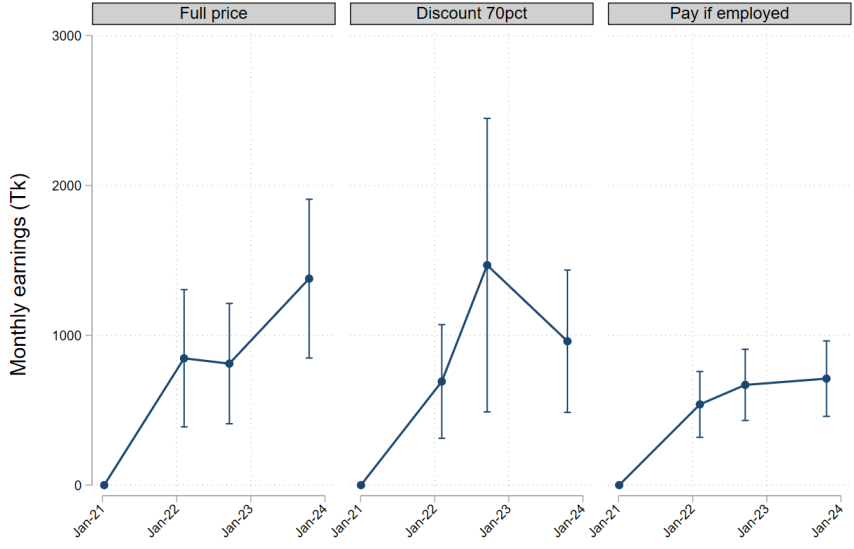


Table: Balance in Characteristics in Full Price vs. Pay-if-employed - Enrolled in Training

	(1) Full Price		(2) Pay-if-employed		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	33	20.939 (1.032)	528	22.970 (0.270)	561	-2.030*
Secondary education	33	0.212 (0.072)	528	0.225 (0.018)	561	-0.013
HH income p.c.	33	4289.364 (359.484)	528	3890.172 (90.311)	561	399.192
Female	33	0.848 (0.063)	528	0.661 (0.021)	561	0.188**
Children	33	0.515 (0.145)	528	0.672 (0.043)	561	-0.157
Employed	33	0.121 (0.058)	528	0.250 (0.019)	561	-0.129*
Reservation wage	19	10168.421 (1670.348)	373	9279.893 (330.736)	392	888.528
Work experience	33	15.303 (7.643)	528	23.801 (1.981)	561	-8.498
Training experience	33	8.030 (4.192)	528	37.428 (6.474)	561	-29.398
Predicted employment	33	0.268 (0.027)	528	0.358 (0.008)	561	-0.090***

* $p<0.1$, ** $p<0.05$, *** $p<0.01$..

Table: Balance in Characteristics in Full Price vs. 70 pct Discount - Enrolled in Training

	(1) Full Price		(2) 70 pct discount		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	33	20.939 (1.032)	229	24.725 (0.459)	262	-3.785***
Secondary education	33	0.212 (0.072)	229	0.231 (0.028)	262	-0.019
HH income p.c.	33	4289.364 (359.484)	229	3564.010 (114.004)	262	725.354**
Female	33	0.848 (0.063)	229	0.777 (0.028)	262	0.071
Children	33	0.515 (0.145)	229	0.956 (0.074)	262	-0.441**
Employed	33	0.121 (0.058)	229	0.253 (0.029)	262	-0.132*
Reservation wage	19	10168.421 (1670.348)	162	8550.926 (385.417)	181	1617.495
Work experience	33	15.303 (7.643)	229	27.293 (3.324)	262	-11.990
Training experience	33	8.030 (4.192)	229	30.122 (10.105)	262	-22.092
Predicted employment	33	0.268 (0.027)	229	0.331 (0.012)	262	-0.063*

* $p<0.1$, ** $p<0.05$, *** $p<0.01$..

Table: Balance in Characteristics by Interest in Training - Full Price vs. Pay-if-employed

	(1) Pay-if-employed only		(2) Full Price and pay-if-employed		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	7657	24.969 (0.079)	5068	24.834 (0.094)	12725	0.135
Secondary education	7657	0.188 (0.004)	5068	0.204 (0.006)	12725	-0.016**
HH income p.c.	7657	3625.154 (21.866)	5068	4255.967 (28.988)	12725	-630.813***
Female	7657	0.797 (0.005)	5068	0.734 (0.006)	12725	0.062***
Children	7657	1.029 (0.013)	5068	0.976 (0.015)	12725	0.053***
Employed	7657	0.272 (0.005)	5068	0.275 (0.006)	12725	-0.003
Reservation wage	5253	7998.784 (73.706)	3223	9072.954 (107.784)	8476	-1074.170***
Work experience	7657	32.545 (0.622)	5068	32.166 (0.780)	12725	0.378
Training experience	7657	32.993 (2.178)	5068	32.008 (3.283)	12725	0.985
Predicted employment	7657	0.338 (0.002)	5068	0.353 (0.003)	12725	-0.015***

* $p<0.1$, ** $p<0.05$, *** $p<0.01$. Sample includes all households surveyed at baseline who are interested in training under pay-if-employed.

Back

Table: Balance in Characteristics in Full Price vs. Pay-if-employed - BRAC Sample

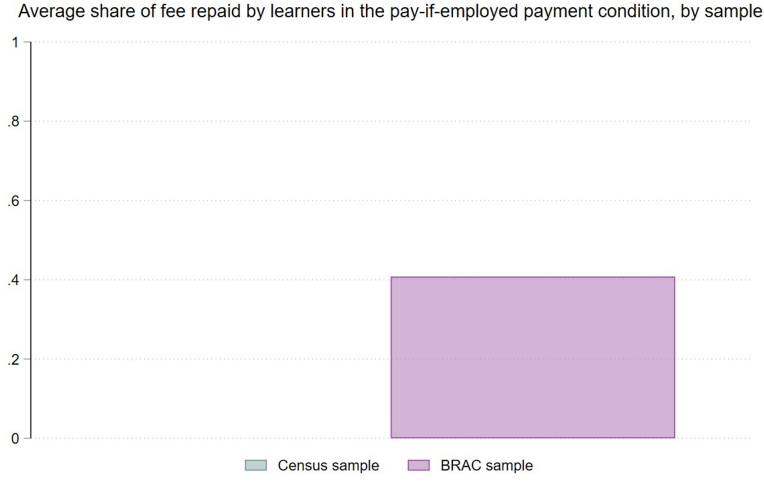
	(1) Pay-if-employed		(2) Full Price		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	765	22.229 (0.235)	390	23.100 (0.315)	1155	-0.871**
Secondary education	765	0.280 (0.016)	390	0.282 (0.023)	1155	-0.002
HH income p.c.	765	4948.565 (160.001)	390	5842.322 (209.854)	1155	-893.757***
Female	765	0.488 (0.018)	390	0.485 (0.025)	1155	0.003
Children	765	0.333 (0.028)	390	0.321 (0.037)	1155	0.013
Employed	765	0.263 (0.016)	390	0.318 (0.024)	1155	-0.055**
Work experience	765	17.240 (1.383)	390	22.295 (2.279)	1155	-5.055**
Training experience	765	23.366 (2.834)	390	43.721 (6.462)	1155	-20.354***
Reservation wage	765	12467.974 (352.864)	390	15045.641 (507.608)	1155	-2577.667***

* $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Back

Minimal repayment in PIE

Back



Moral hazard in reporting

Table: Impact of Reminder on Reported Earnings and Employment

VARIABLES	(1) Employed	(2) Earnings
Message	-0.05	-1,260.02***
	(0.03)	(439.21)
Observations	6,213	6,213
R-squared	0.074	0.074
Mean	0.284	2062

SE clustered by branch-trade

*p<.05; **p<.01; ***p<.001

Back

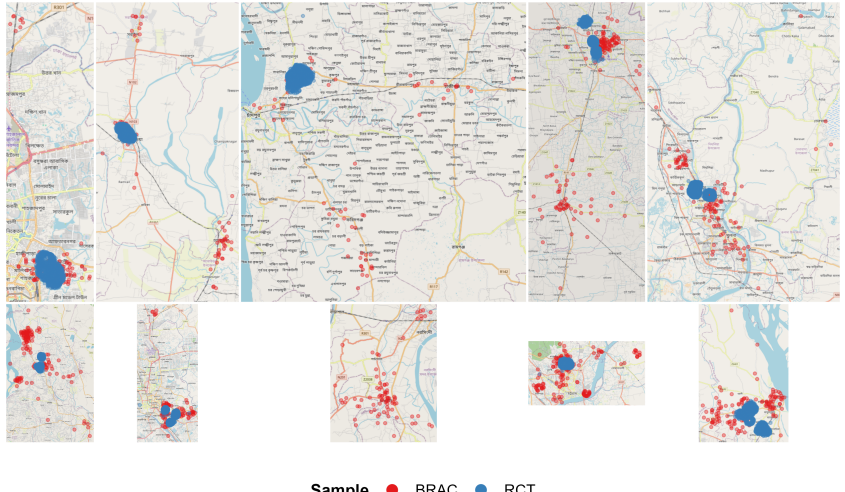
Table: Treatment effects with Willingness to Pay Interactions

	Employed	Weekly work hrs	Monthly earnings
	(1)	(2)	(3)
Panel A: Treatment effects for WTP $\geq$ 100			
Full price	-0.009.	1.223.	0.390.
	(0.029)	(1.125)	(0.327)
Discount 30pct	-0.009.	-1.178.	-0.562.
	(0.040)	(1.498)	(0.456)
Discount 70pct	0.047*	0.603.	0.091.
	(0.027)	(0.942)	(0.279)
Pay if employed	0.023.	1.027.	-0.083.
	(0.028)	(0.862)	(0.238)
Panel B: Main effect for WTP $<$ 100			
WTP $<$ 100	0.024.	0.100.	-0.197.
	(0.027)	(1.010)	(0.286)
Panel C: Discounts $\times$ WTP $<$ 100			
Discount 30pct	-0.071.	0.106.	-0.022.
	(0.067)	(2.219)	(0.408)
Discount 70pct	-0.072**	-0.892.	-0.257.
	(0.031)	(1.209)	(0.350)
Pay if employed	-0.038.	-0.867.	0.135.
	(0.038)	(1.231)	(0.399)
Control mean	0.344	9.749	2.533
Observations	6,675	6,675	6,675
R-squared	0.039	0.064	0.098

Standard errors clustered by branch-trade. Earnings are measured in 1000 Tk.
Full price treatment was only available to participants with WTP $\geq$ 100.
*** p<0.01, ** p<0.05, * p<0.1

Back

Figure: Coordinates of BRAC and Experimental Sample Respondents (Displaced)



Back

Weighted Hours Treatment Effects

Table: Hours Treatment Effects

	Unweighted (1)	Variable Wts (2)	Takeup Wts (3)
Full price	-0.597	1.543	1.239
	(0.898)	(1.715)	(1.516)
Discount 30pct	-1.489	-2.935	-2.781
	(1.348)	(2.268)	(2.171)
Discount 70pct	1.000	3.691***	3.089***
	(0.693)	(0.989)	(1.071)
p-value for equality of treatment effects:			
	0.168	0.022	0.047
Control mean	11.592	11.592	11.592
Observations	4,246	4,246	4,246
R-squared	0.303	0.307	0.303
Rerandomized Controls	Yes	Yes	Yes

Outcome is mean work hours at endline. Standard errors are clustered by branch-trade. All regressions include `ib1.ctreat` (Full price as the omitted category) and rerandomized balance controls. Control mean is unweighted (`invite==0`). *** p$¡$0.01, ** p$¡$0.05, * p$¡$0.1.

Back

Weighted Earnings Treatment Effects

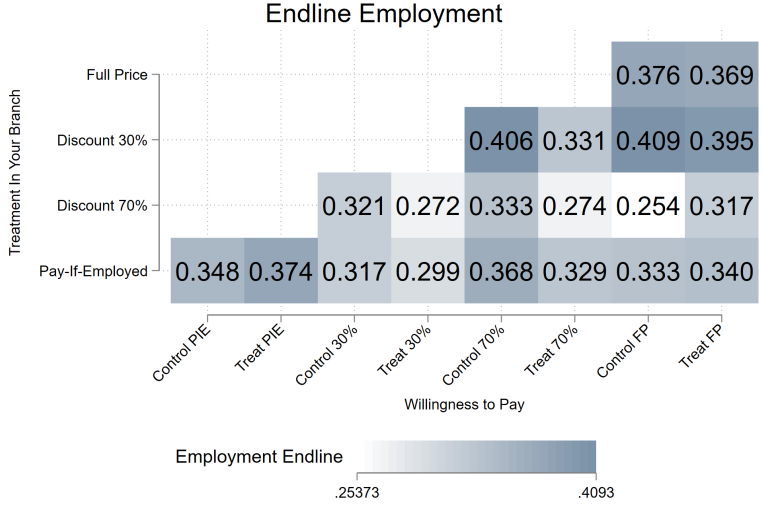
Table: Earnings Treatment Effects

	Unweighted (1)	Variable Wts (2)	Takeup Wts (3)
Full price	-253.138	262.160	187.384
	(275.482)	(594.922)	(551.590)
Discount 30pct	-403.123	-948.714	-713.638
	(310.166)	(582.731)	(621.505)
Discount 70pct	163.655	780.290	692.692
	(171.010)	(559.738)	(464.604)
p-value for equality of treatment effects:			
	0.206	0.086	0.188
Control mean	3117.483	3117.483	3117.483
Observations	4,246	4,246	4,246
R-squared	0.389	0.363	0.351
Rerandomized Controls	Yes	Yes	Yes

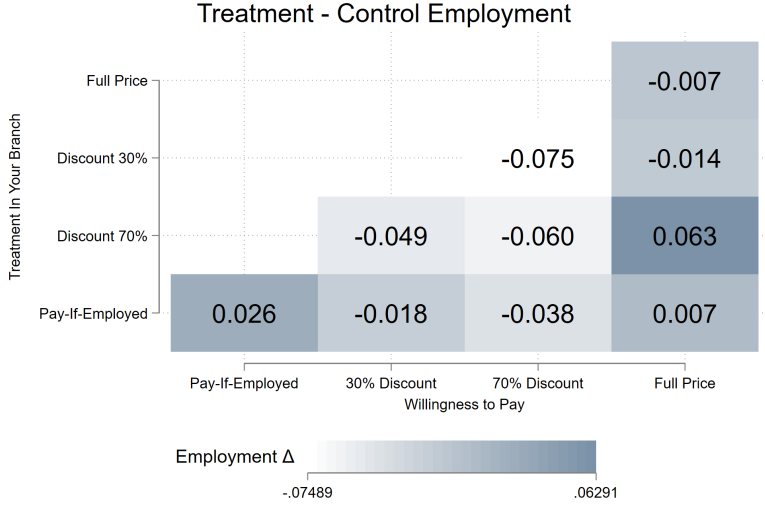
Outcome is earnings at endline (in Tk). Standard errors are clustered by branch-trade. All regressions include `ib1.ctreat` (Full price as the omitted category) and rerandomized balance controls. Control mean is unweighted (`invite==0`). *** p$¡$0.01, ** p$¡$0.05, * p$¡$0.1.

Back

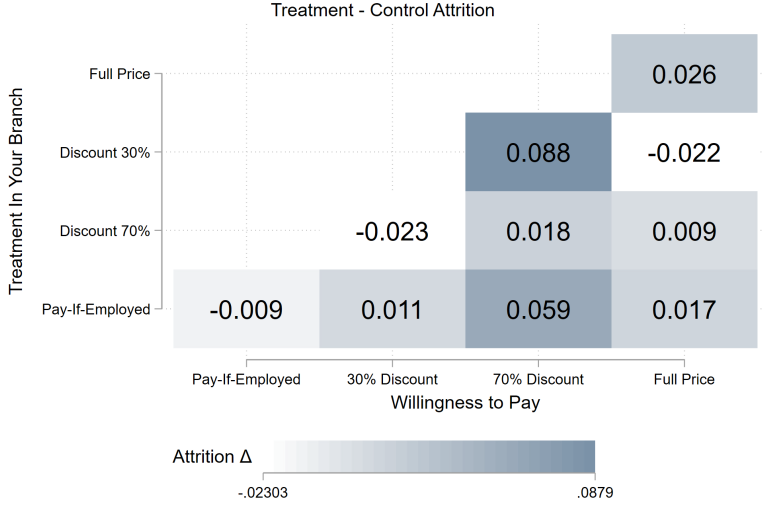
Endline Employment Heatmap - Version 1



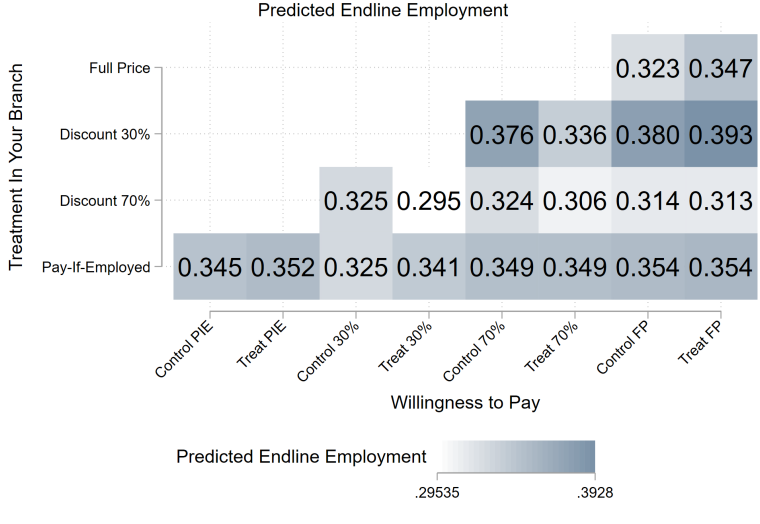
Endline Employment Heatmap - Version 2



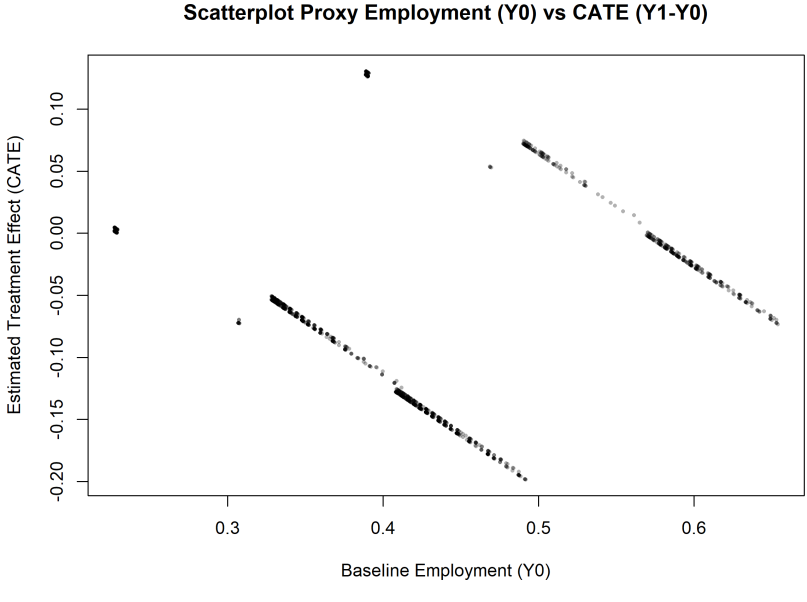
Attrition


Treatment - Control Attrition
Treatment In Your Branch
Full Price
Discount 30%
Discount 70%
Pay-If-Employed
0.026
0.088
-0.022
-0.023
0.018
0.009
-0.009
0.011
0.059
0.017
Pay-If-Employed
30% Discount
70% Discount
Full Price
Willingness to Pay
Attrition Δ
-.02303
.0879

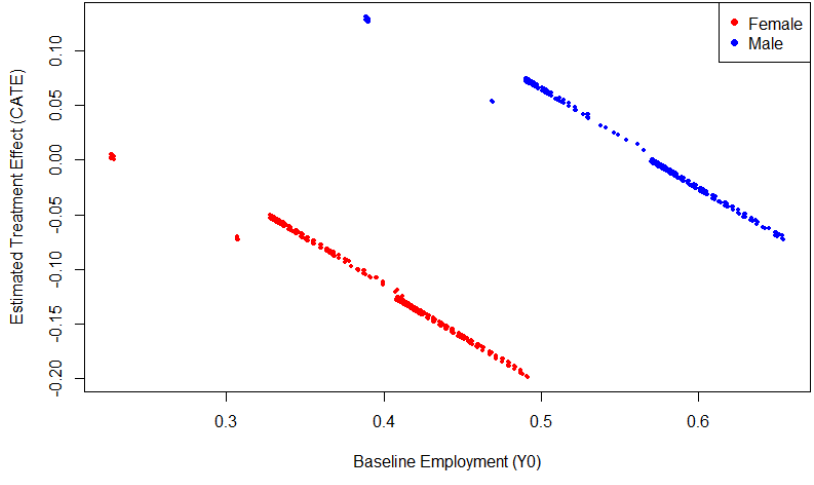
Attrition


Predicted Endline Employment
Treatment In Your Branch
Full Price
Discount 30%
Discount 70%
Pay-If-Employed
0.323 0.347
0.376 0.336 0.380 0.393
0.325 0.295 0.324 0.306 0.314 0.313
0.345 0.352 0.325 0.341 0.349 0.349 0.354 0.354
Control PIE
Treat PIE
Control 30%
Treat 30%
Control 70%
Treat 70%
Control FP
Treat FP
Willingness to Pay
Predicted Endline Employment
.29535
.3928

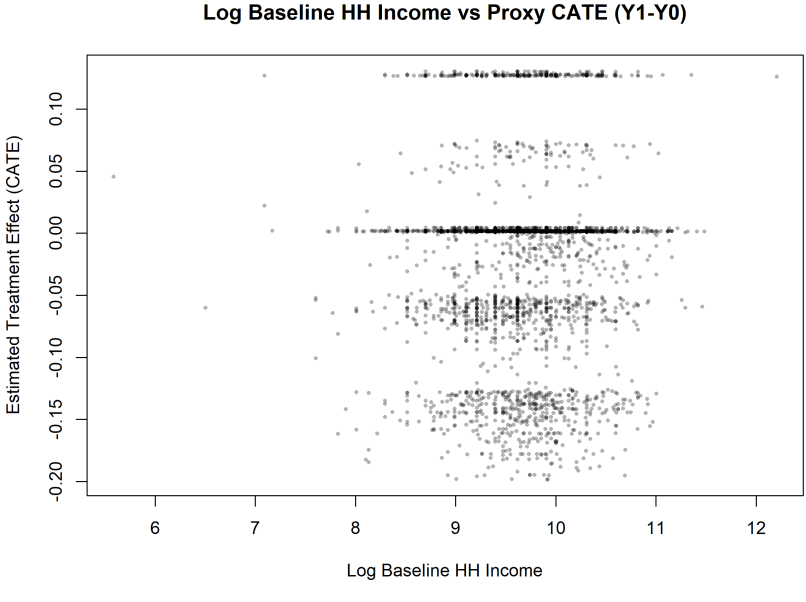
Proxy Conditional Average Treatment Effects



Proxy Conditional Average Treatment Effects by Gender

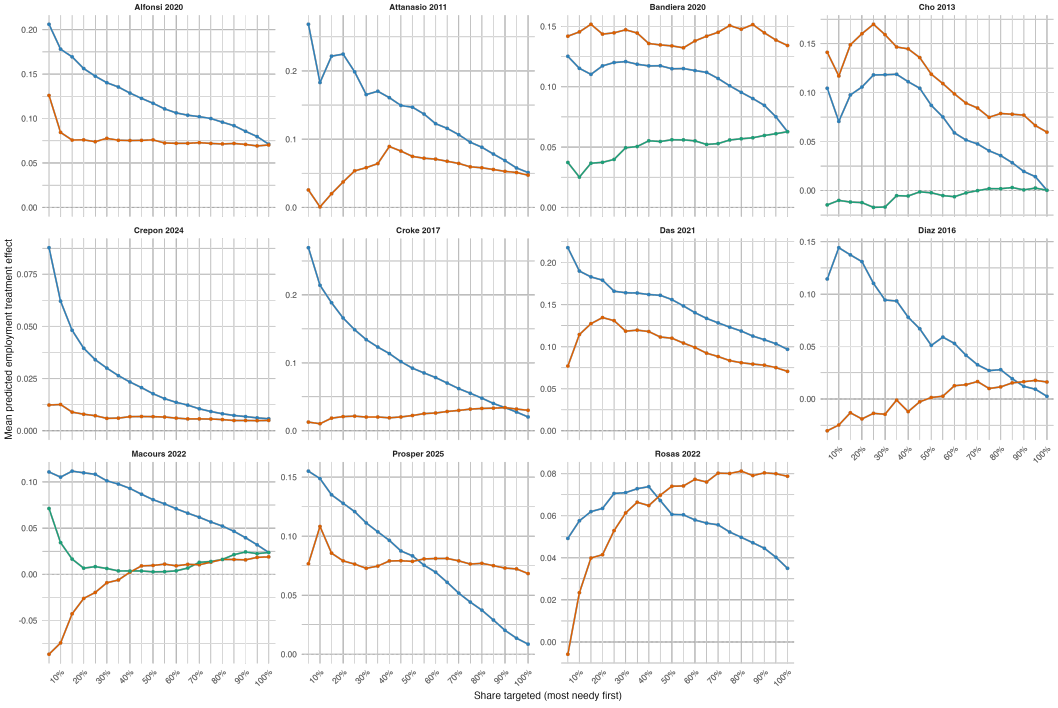


Proxy Conditional Average Treatment Effects over Baseline HH Income



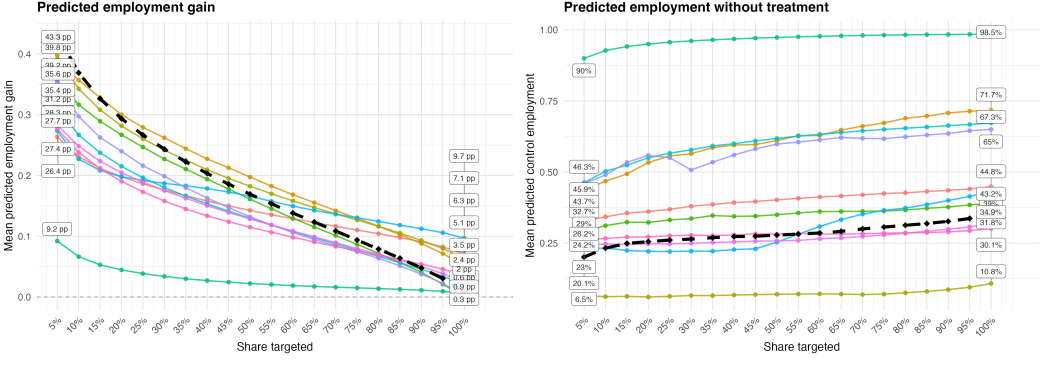
Needs-based targeting — by study

Individuals ranked by baseline need; employment gain among selected



Employment targeting - by study

Individuals ranked by predicted employment treatment effect; top-ranked share selected

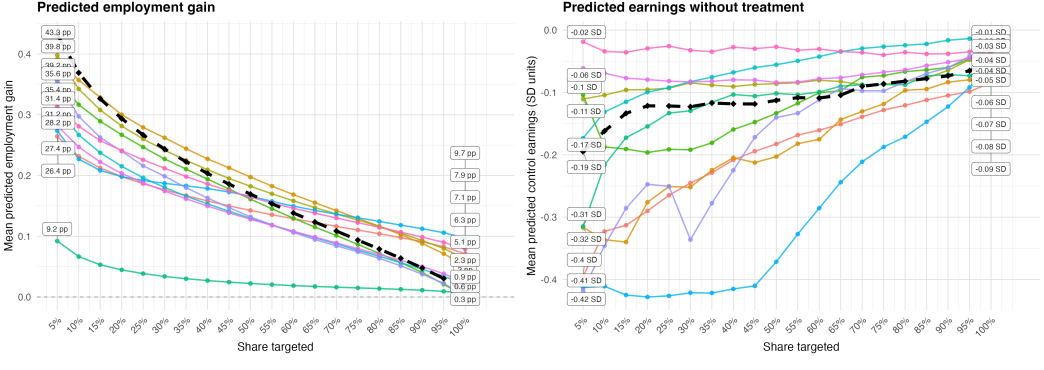


Notes: Individuals are ranked within study by the relevant GenericML predicted treatment effect. Each point reports the mean among the top-ranked share of individuals. Dashed black line: Prosper 2025.

Back

Employment targeting and baseline earnings - by study

Individuals ranked by predicted employment treatment effect; top-ranked share selected

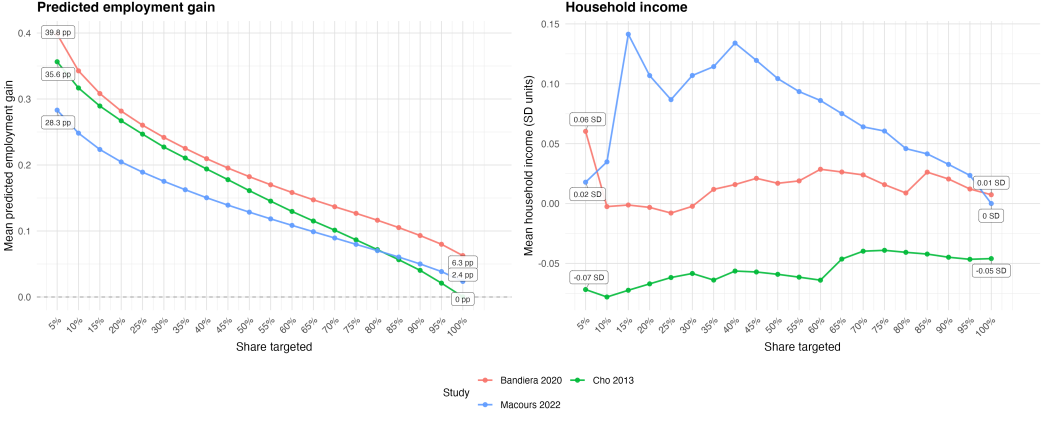


Notes: Individuals are ranked within study by the relevant GenericML predicted treatment effect. Each point reports the mean among the top-ranked share of individuals. Earnings outcomes are standardized within study. Dashed black line: Prosper 2025.

Back

Employment targeting and household income - by study

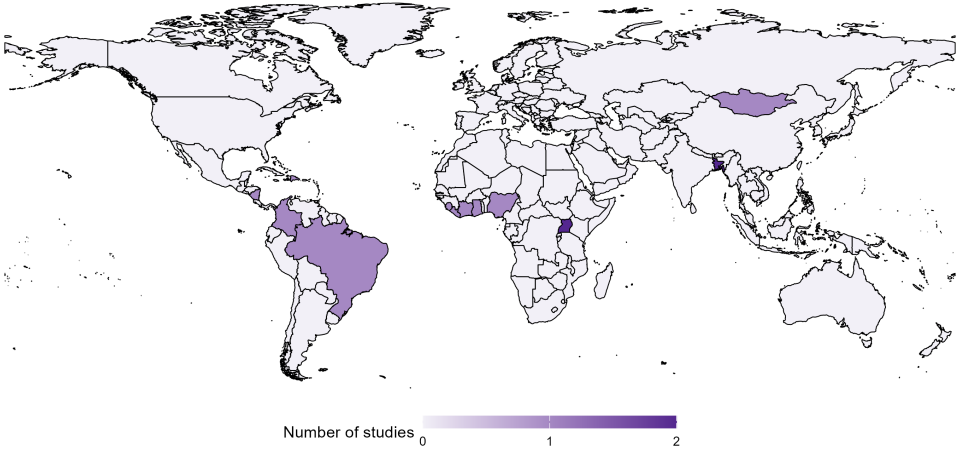
Individuals ranked by predicted employment treatment effect; top-ranked share selected



Notes: Individuals are ranked within study by the relevant GenericML predicted treatment effect. Each point reports the mean among the top-ranked share of individuals. Household income is standardized within study.

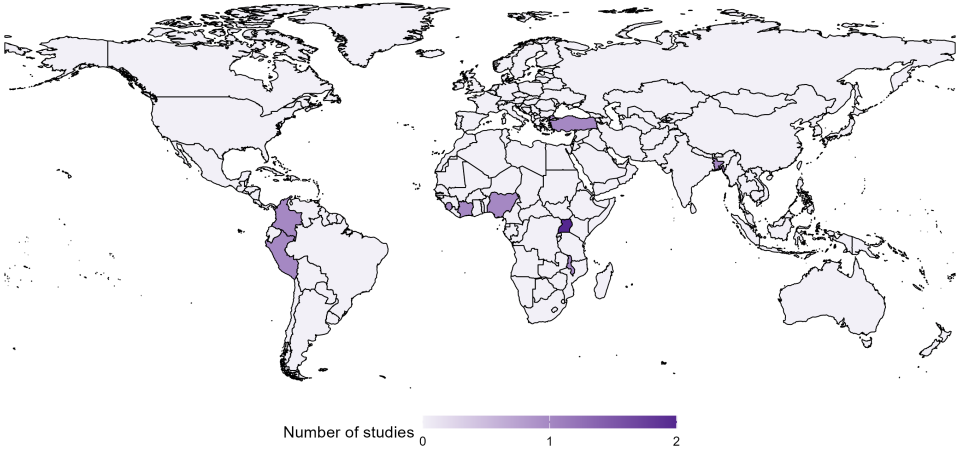
Back

Countries represented in Agarwal and Mani (2024) review of 14 studies



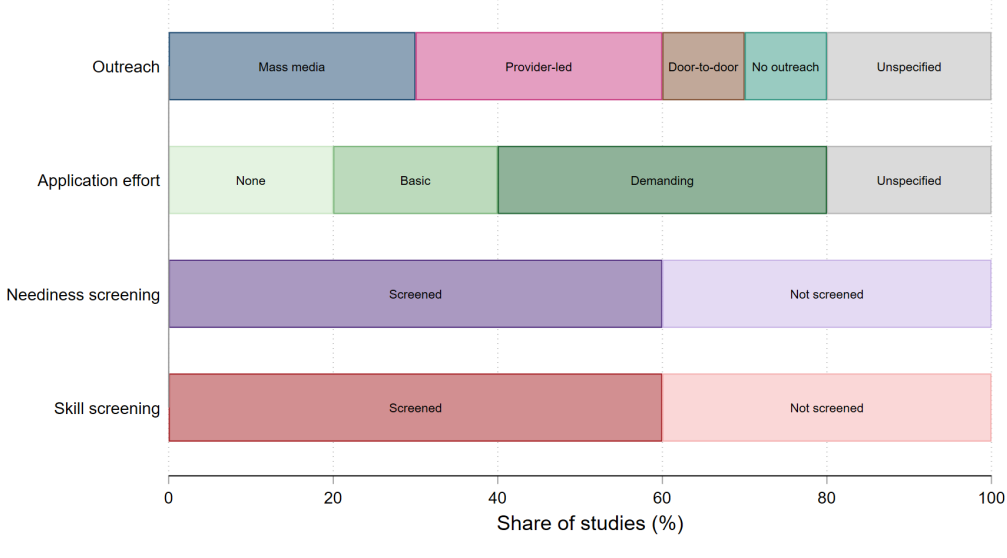
Back

Countries represented in reanalysis of 10 training RCT evaluations



Back

Recruitment and screening in re-analysis of 10 training-RCT evaluations



Back

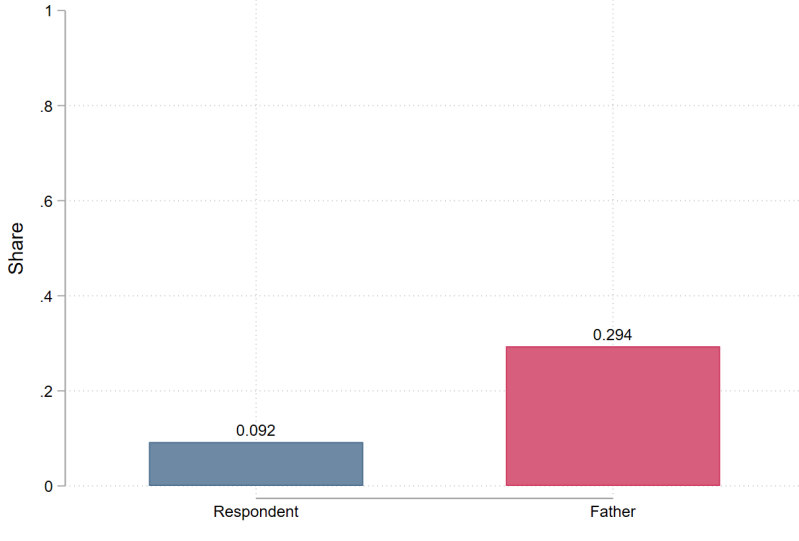
Table: Effect of Subsidies on Enrollment, by Poverty

	(1)	(2)
Constant	0.145	0.170
	(0.053)	(0.049)
Income pc	-0.000	
	(0.000)	
Poverty		-0.010
		(0.019)
30% discount	0.012	0.012
	(0.028)	(0.016)
30% discount × Income pc	0.000	
	(0.000)	
30% discount × Poverty		-0.004
		(0.023)
70% discount	0.134***	0.087***
	(0.037)	(0.021)
70% discount × Income pc	-0.000*	
	(0.000)	
70% discount × Poverty		0.010
		(0.030)
R^2	0.041	0.040
Observations	4290	4290

Sample includes households invited to training under 3 treatments. All regressions control for trade and rerandomisation variables. Poverty is an indicator for HH income per capita below $3/day (2021 PPP). SE clustered at branch-trade. * p<0.1, ** p<0.05, *** p<0.01.

Back

Figure: Share of Sample Working in Agriculture



Econometric specification

Back

$$y_{ibtd} = \alpha + \sum_d \beta_d I(discount = d)_{bt} * offer_i + \sum_d \gamma_d I(discount = d)_{bt} + \theta_t + X_i + u_{ibt} \quad (1)$$

y_{ibt} is an outcome for individual i, in branch b, who is most interested in trade t.

$I(discount = d)_{bt}$ identifies clusters assigned to discount d ($d \in \{0, .3, .7\}$). $offer_i$ identifies individuals invited for treatment.

θ_t is a trade dummy. X_i are re-randomization baseline covariates.

We collected data at baseline (December 2020), midline (January and September 2022) and endline (September 2023).

Balance and Attrition

Back

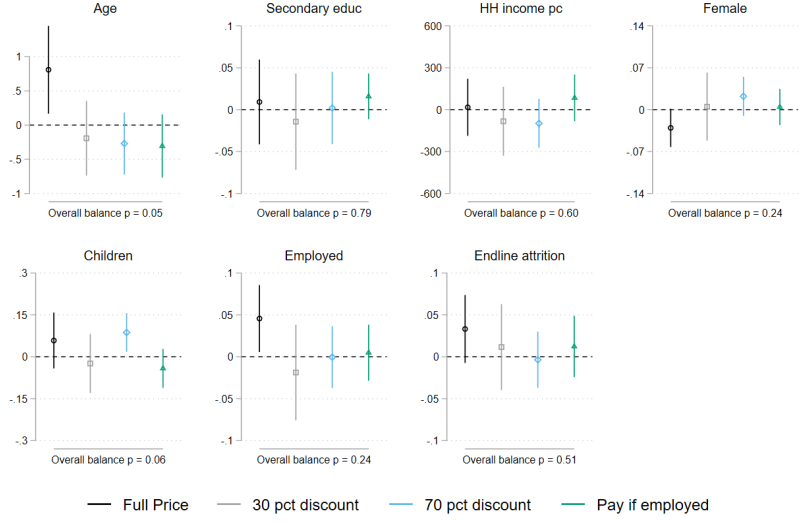
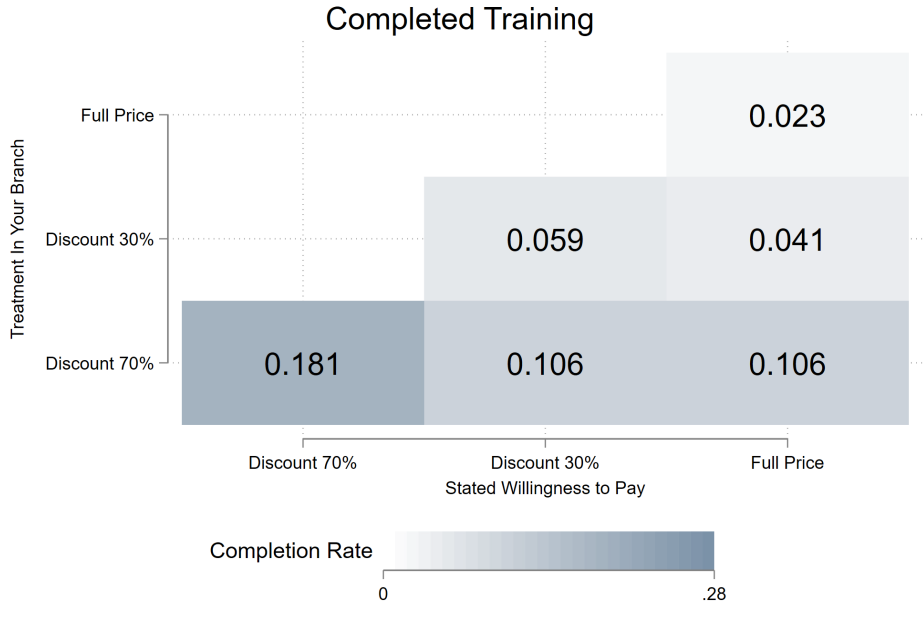


Table Tables by treatment

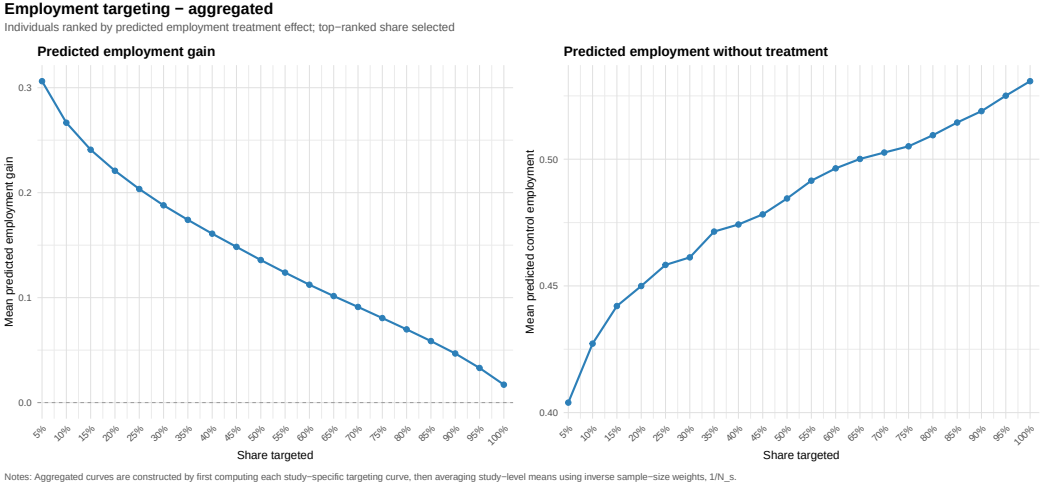
Both intensive and extensive margin effects

Back



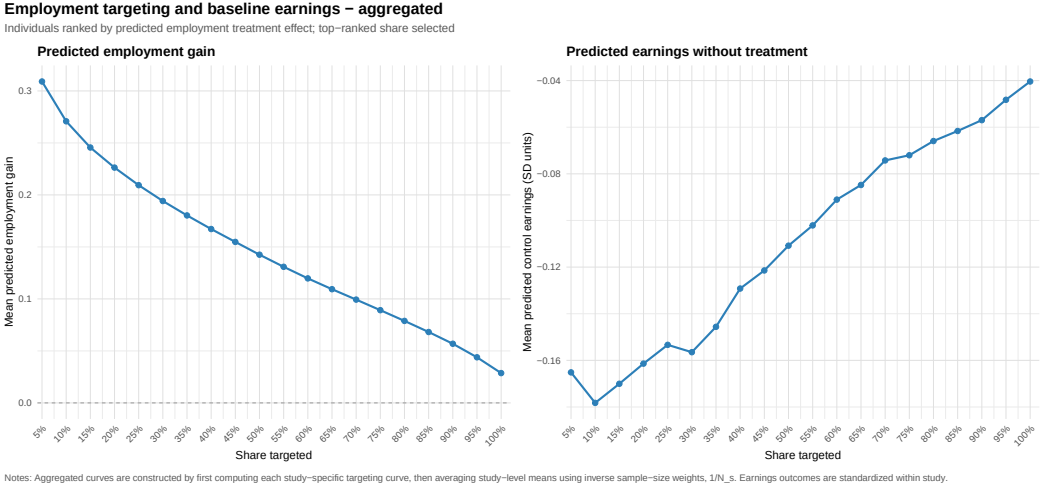
Those with the highest predicted gains from training have high predicted neediness in terms of employment

Back



By study

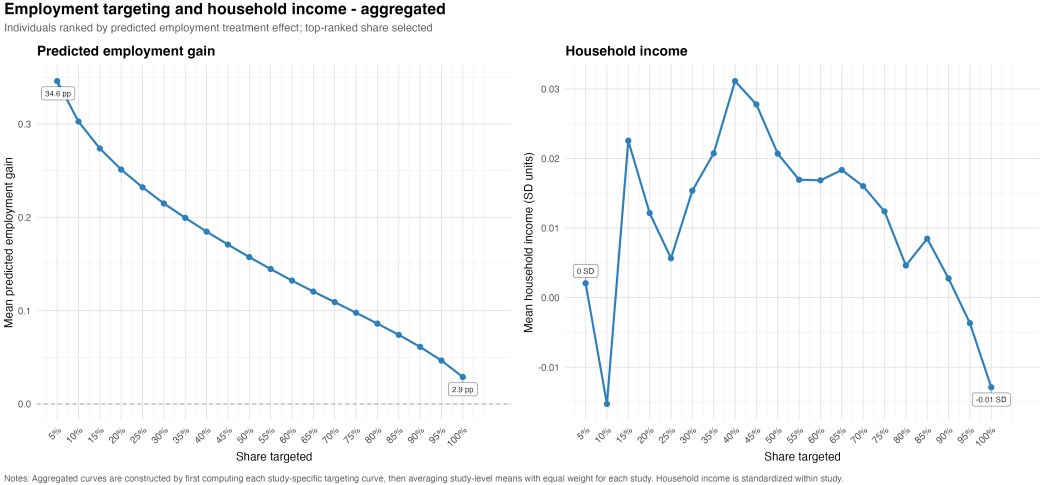
And in terms of earnings



Notes: Aggregated curves are constructed by first computing each study−specific targeting curve, then averaging study−level means using inverse sample−size weights, 1/N_s. Earnings outcomes are standardized within study.

By study

And are similarly poor



This is particularly true for the poor

Back

Table: Weighted Treatment Effects on Labor Market Outcomes

	Unweighted			Takeup Weights		
	Enrolled	Employed	Earnings	Enrolled	Employed	Earnings
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Poor (below 3.00 USD/day)						
Full price	0.033	-0.021	-295.768	0.056*	-0.016	343.636
	(0.028)	(0.056)	(574.793)	(0.032)	(0.079)	(1048.690)
Discount 30pct	0.067***	-0.085	-470.987	0.098***	-0.059	-891.535
	(0.025)	(0.064)	(635.459)	(0.035)	(0.064)	(976.972)
Discount 70pct	0.162***	0.090**	242.273	0.255***	0.187***	1351.362
	(0.034)	(0.040)	(422.588)	(0.048)	(0.057)	(968.712)
Control mean	0.000	0.332	2659.185	0.000	0.404	3784.082
Observations	1,074	1,074	1,074	1,074	1,074	1,074
Panel B: Non-poor (above 3.00 USD/day)						
Full price	0.032*	-0.044	-219.909	0.029*	-0.017	283.769
	(0.018)	(0.037)	(313.100)	(0.016)	(0.052)	(581.515)
Discount 30pct	0.059***	-0.001	-443.761	0.065***	-0.024	-768.828
	(0.015)	(0.029)	(374.486)	(0.016)	(0.031)	(728.307)
Discount 70pct	0.137***	-0.016	139.967	0.157***	0.007	451.061
	(0.021)	(0.024)	(209.014)	(0.033)	(0.033)	(616.383)
Control mean	0.000	0.363	3266.474	0.000	0.455	4983.821
Observations	3,172	3,172	3,172	3,172	3,172	3,172
p-value for equality of treatment across panels:						
	0.901	0.524	0.955	0.226	0.066	0.972
Rerandomized Controls	Yes	Yes	Yes	Yes	Yes	Yes

Sample split by baseline household income per capita relative to the $3.00/day (2021 PPP) extreme poverty line. Standard errors are clustered by branch-trade. Control variables are rerandomized balance controls, a dummy variable for baseline student status, preferred trade, baseline earnings, self-employed status at baseline, self-employed status in Nov. 2019, and aspired earnings. Earnings are measured in Tk. Individual food expenditure is the monthly equivalent of the response to: In the past 7 days, how many days have you eaten the following food items listed? How much would it cost to buy all of the food you consumed from this category this week at a grocery store in taka? Household non-food expenditure is the sum of monthly expenditure on non-food non-durables and the annual expenditure on durables divided by 12. Earnings and consumption variables are winsorized at the 99th percentile. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Selector CV ranking

Back

- We invited the selectors to consider 20 real, anonymised CVs.
- Each CV was scored on 4 dimensions: likelihood of take-up, expected benefit, expected neediness, likelihood that they would repay their debt if offered PIE.
- Then they would pick 5 CVs out of 20 to invite for Prosper.

Selector CV ranking

Prosper Applicant CV 12

Ideal trade: Tailoring (female)

Personal Information

- Gender: female
- Age: 25
- Marital status: currently or previously married
- Number of children: 1
- Household income per month: 7500 Taka
- Father's occupation type: low skill
- Mother's occupation type: never worked

Education and Work Experience

- Previous work experience: 0 months
- Previous vocational training: no
- Completed SSC/Dakhil: yes

BRAC targets mostly on expected take-up

Table: Relationship between CV Selection and Scores

VARIABLES	(1) CV picked	(2) CV picked	(3) CV picked	(4) CV picked
Enrol score	0.08***			0.09***
	(0.02)			(0.02)
Benefit score		0.06***		-0.00
		(0.02)		(0.02)
Neediness score			0.01	-0.00
			(0.02)	(0.02)
Observations	200	200	200	200
R-squared	0.110	0.045	0.002	0.110
Mean score	3.020	3.355	3.155	
SD score	1.710	1.591	1.607	

*p<.05; **p<.01; ***p<.001